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IOT (and M2M and WoT)

Day 1 - Foundations | 2026

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The course: three days, one story

Day 1 - Foundations

- What IoT is - and why now
- Networking: OSI and TCP/IP
- Wireshark: seeing packets
- Hardware interfaces
- Short-range wireless

TODAY

Day 2 - Technologies

- Architecture and cloud
- LPWAN, cellular, satellite
- LoRa and LoRaWAN
- Zigbee, Thread, Matter
- MQTT and brokers
- Grafana dashboards
- MakeCode warm-up

Day 3 - Hands-on lab

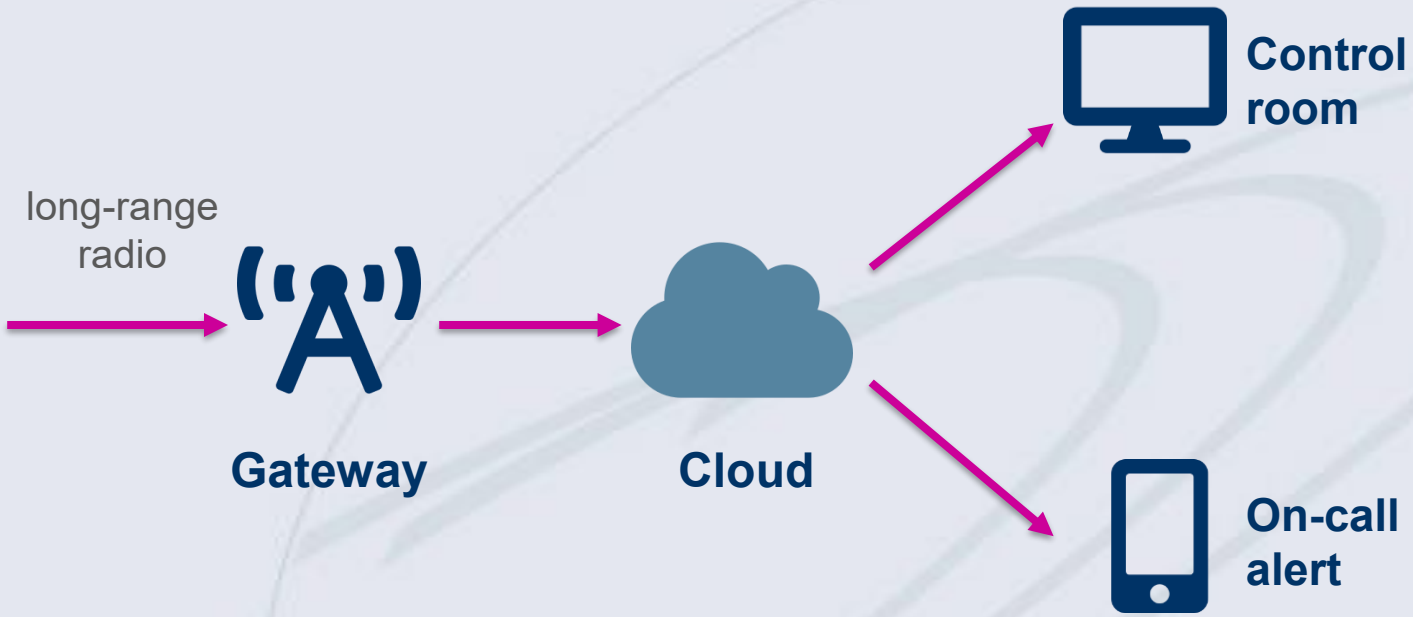
- AWS IoT Core
- Certificates and mTLS
- micro:bit to the cloud
- MQTT + Wireshark + Grafana
- Cloud-to-device control
- Troubleshooting challenges

Our running scenario: Remote Site Watch

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Unmanned remote site

- Temperature
- Door opened
- Vibration / tamper
- Water level
- Battery level

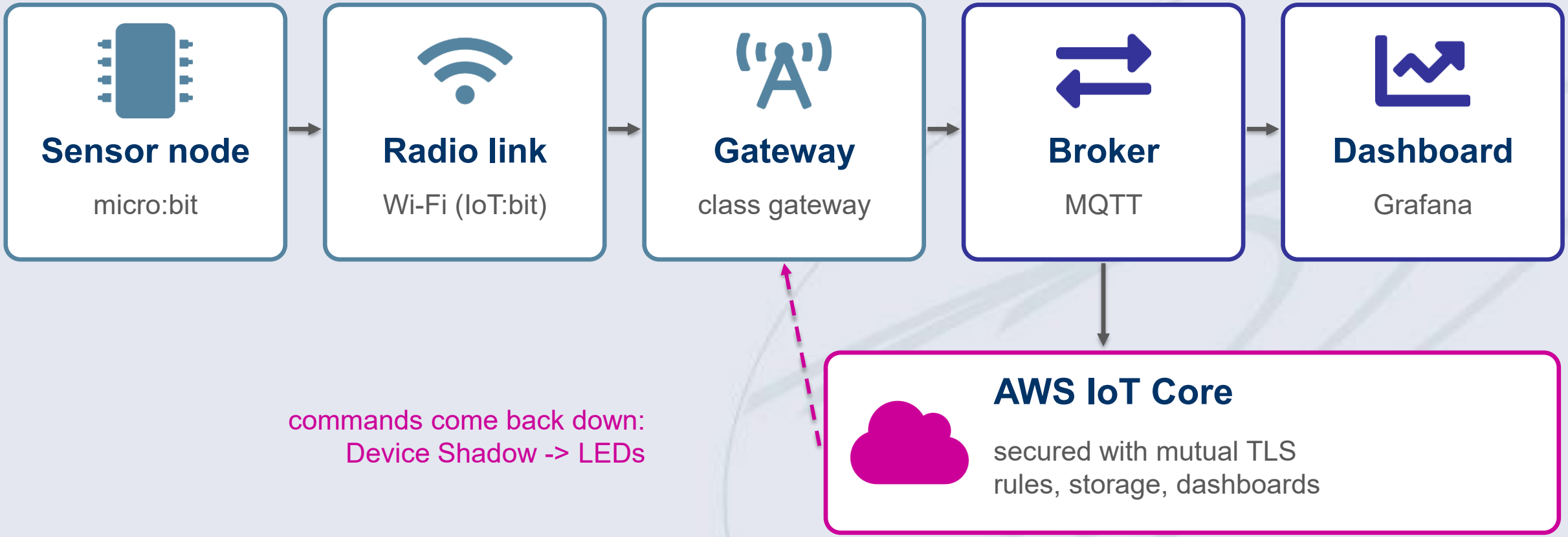


Day 1: the parts - sensors, interfaces, radios, networks

Day 2: the system - LoRaWAN, MQTT, dashboards

Day 3: we build a mini version - micro:bit to AWS IoT

The pipeline we build - one step per day



Day 1 - understand each part Day 2 - messaging, dashboards Day 3 - build it end to end

Day 1 - Foundations: today's plan

1 

Introduction to IoT

uses, challenges, standards

2 

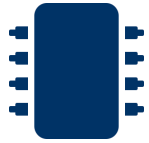
Networking basics

OSI and TCP/IP - layers 4, 5, 7

3 

Wireshark

live demo + 5-minute try

4 

Hardware interfaces

GPIO, UART, I2C, SPI

5 

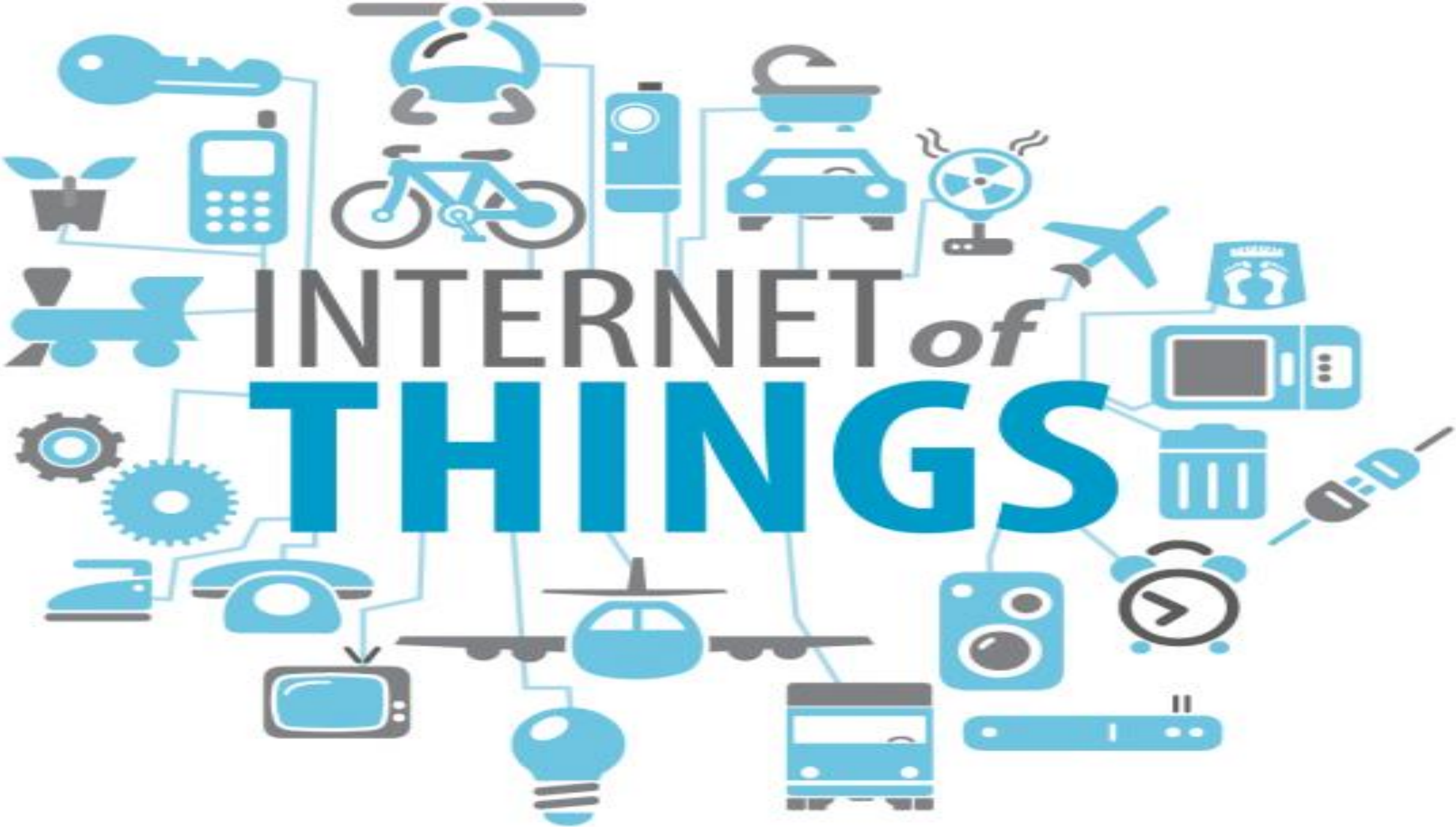
Short-range wireless

Wi-Fi, Bluetooth, NFC, UWB

6 

Summary and quiz

what to remember


INTERNET of
THINGS

What is M2M / IoT / WoT?

- Machine-to-Machine (M2M) communications represent technological solutions and deployments allowing Machines, Devices or Objects to communicate with each other, w/o any human interventions

- IoT (Internet of things) - objects that communicate, send information (sensors) and/or receive controls

- WoT - Object that communicate
  via the Web, Cloud



IoT ? (Some Industry Definitions)

- *A network connecting (either wired or wireless) devices, or 'things', that is characterized by autonomous provisioning, management, and monitoring. The IoT is innately analytical and integrated* **(IDC)**

- *IoT is the next evolution of the Internet, connecting the unconnected people, processes, data, and things in your business today* **(Cisco)**

- *IoT devices as those capable of two-way data transmission (excluding passive sensors and RFID tags). It includes connections using multiple communication methods such as cellular, short range and others.* **(GSMA)**

- *Sensors & actuators connected by networks to computing systems. These systems can monitor or manage the health and actions of connected objects and machines. Connected sensors can also monitor the natural world, people, and animal"* **(McKinsey)**

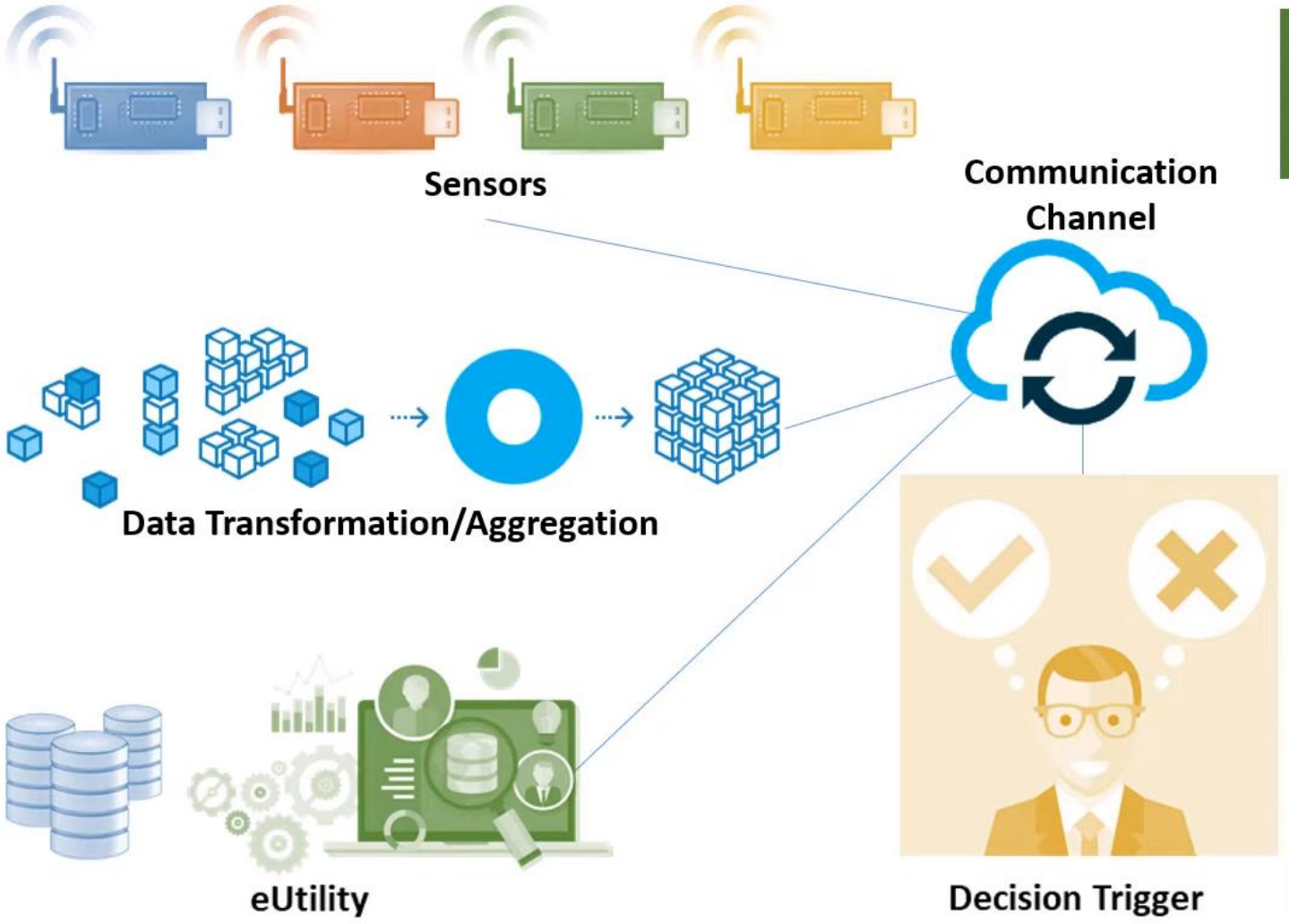
IoT? (IEEE)

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"An IoT system is a network of networks where, typically, a massive number of objects, things, sensors or devices are connected through communications and information infrastructure to provide value-added services via intelligent data processing and management for different applications (e.g. smart cities, smart health, smart grid, smart home, smart transportation, and smart shopping)."

-- IEEE Internet of Things Journal

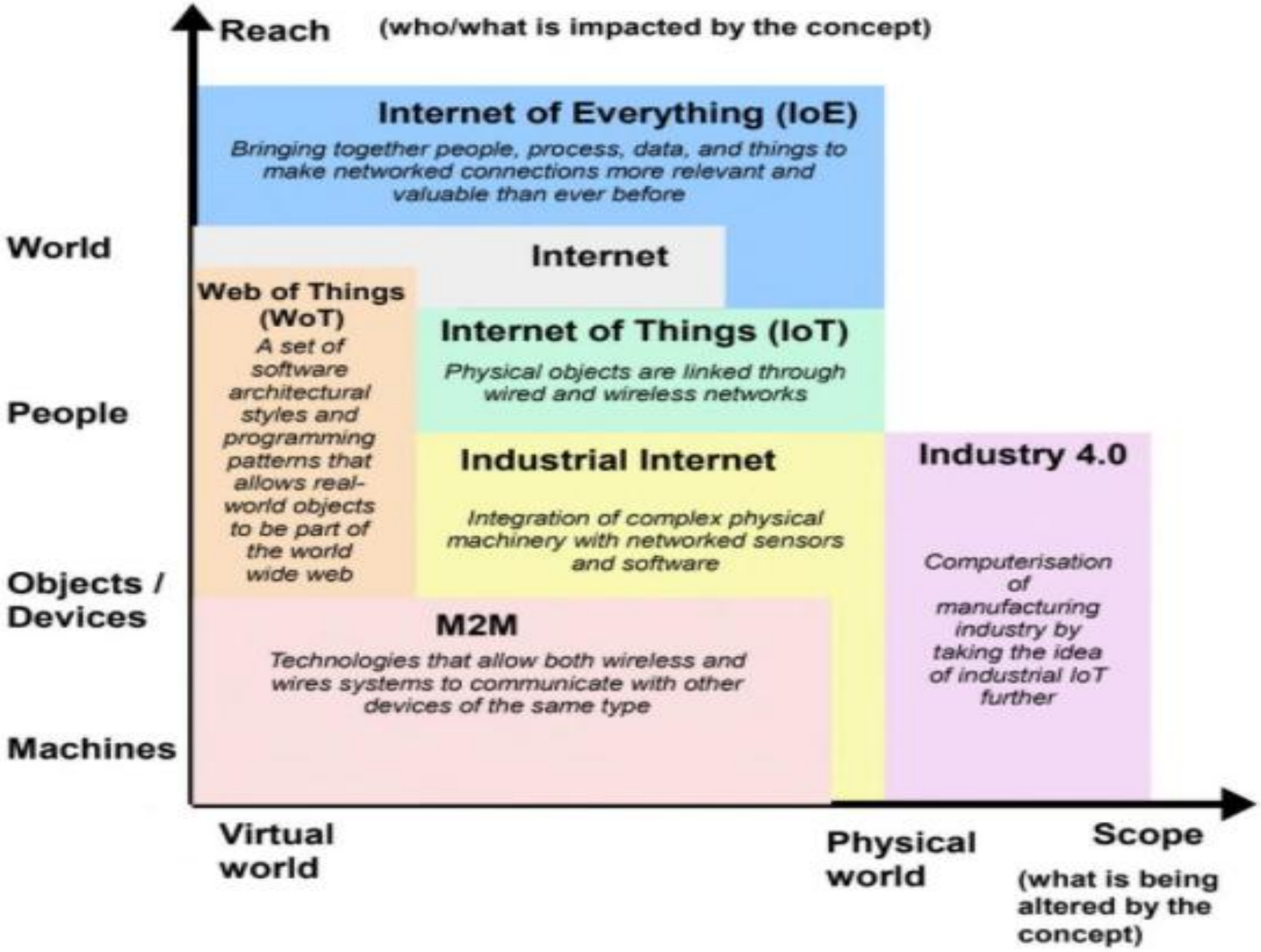
The 5 parts of IoT



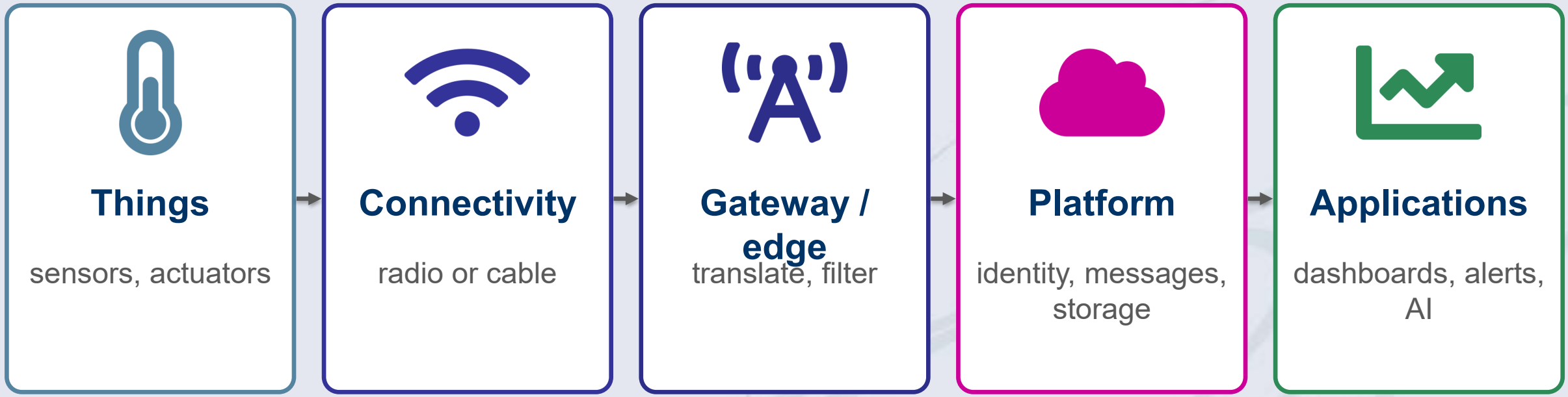
Five Knowledge Areas of IoT

Sensors
Aggregators
Communication Channels
eUtility
Decision Trigger

CWNP Official: The *Internet of Things (IoT)* is the application of connected objects (cyber-physical systems and other objects) that use Internet technologies for automation, reporting, and human interaction.


Reach
(who/what is impacted by the concept)
Internet of Everything (IoE)
Bringing together people, process, data, and things to make networked connections more relevant and valuable than ever before
World
Internet
Web of Things (WoT)
A set of software architectural styles and programming patterns that allows real-world objects to be part of the world wide web
Internet of Things (IoT)
Physical objects are linked through wired and wireless networks
People
Industrial Internet
Integration of complex physical machinery with networked sensors and software
Industry 4.0
Computerisation of manufacturing industry by taking the idea of industrial IoT further
Objects / Devices
M2M
Technologies that allow both wireless and wires systems to communicate with other devices of the same type
Machines
Virtual world
Physical world
Scope
(what is being altered by the concept)

The building blocks of every IoT system



Security and device management run through every block

Our site: door sensor -> LoRa -> mast gateway -> cloud broker -> dashboard + SMS alert

Sensors sense, actuators act

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Sensors: world -> data

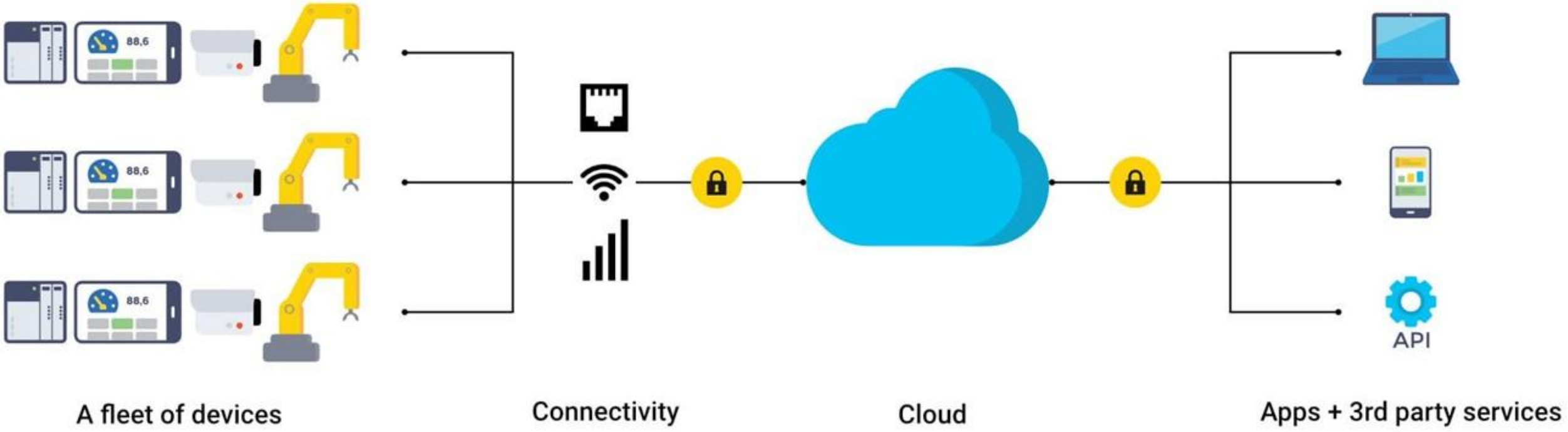
- Temperature
- Motion
- Location
- Light, sound
- Level, flow
- Camera

Actuators: data -> world

- Lights, displays
- Relays
- Motors, valves
- Sirens
- Locks
- Heating

The loop: sense -> send -> decide -> act.
Decide in the cloud, on the gateway, or on the device (edge AI).

Industrial Internet Of Things (IIoT)



INDUSTRY 4.0

MANUFACTURING AND
DIGITAL INNOVATION

ENGINEERING/ PROTOTYPING

ENGINEERING AND
3D PRINTING

Results Engineering™

DELIVERY/ TRACKING

QUICK DELIVERY
AND SHIPMENT TRACKING



DESIGN INNOVATION

INNOVATIVE PRODUCTS
AND NEW TECHNOLOGIES

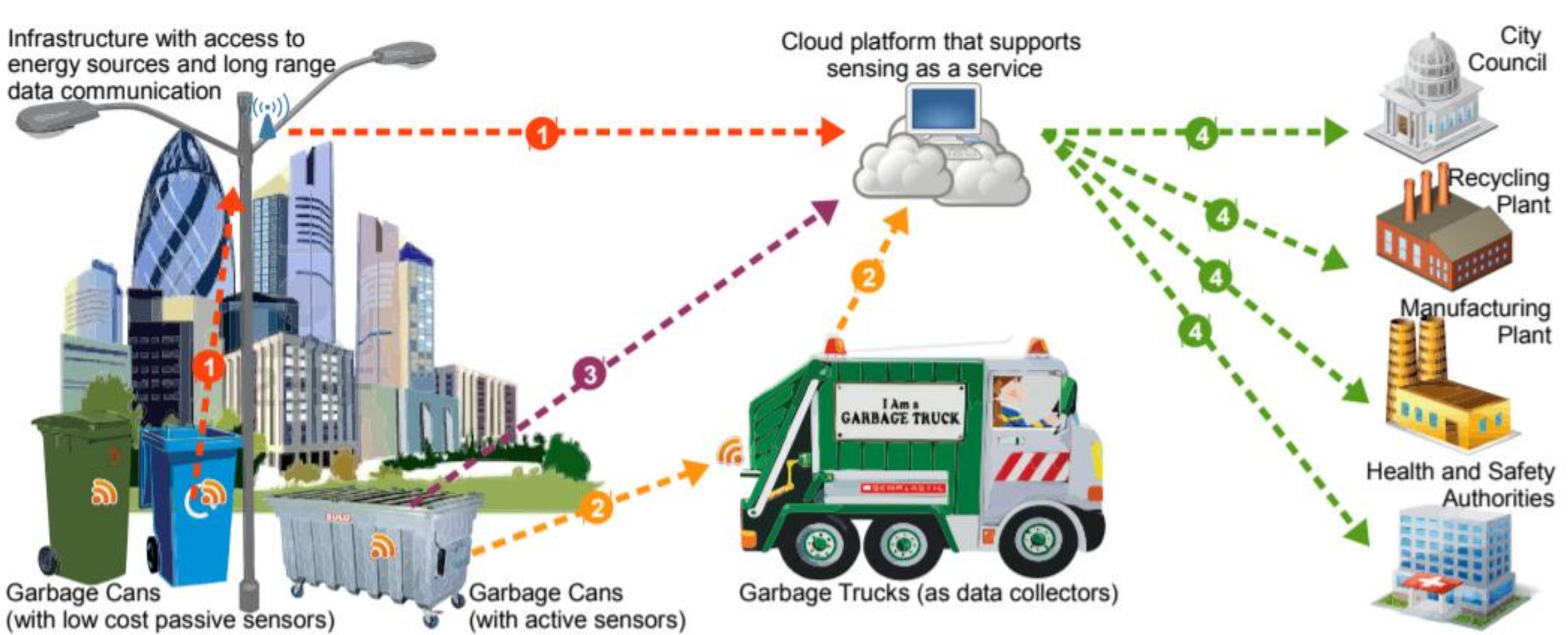
PRODUCTION LINE

AUTOMATED
PRODUCTION LINE

DISTRIBUTION/ ASSISTANCE

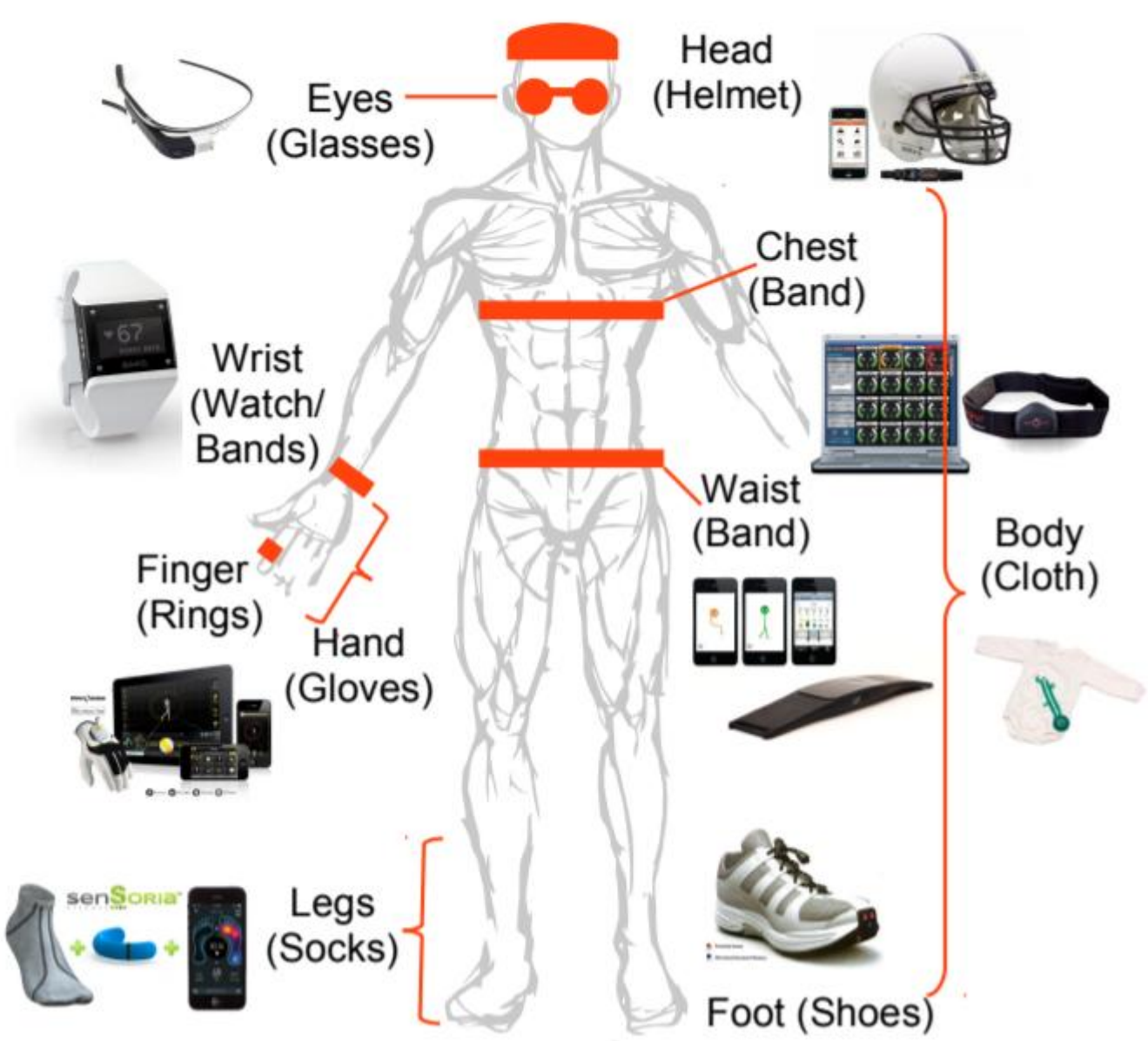
RETAIL
CUSTOMER SUPPORT

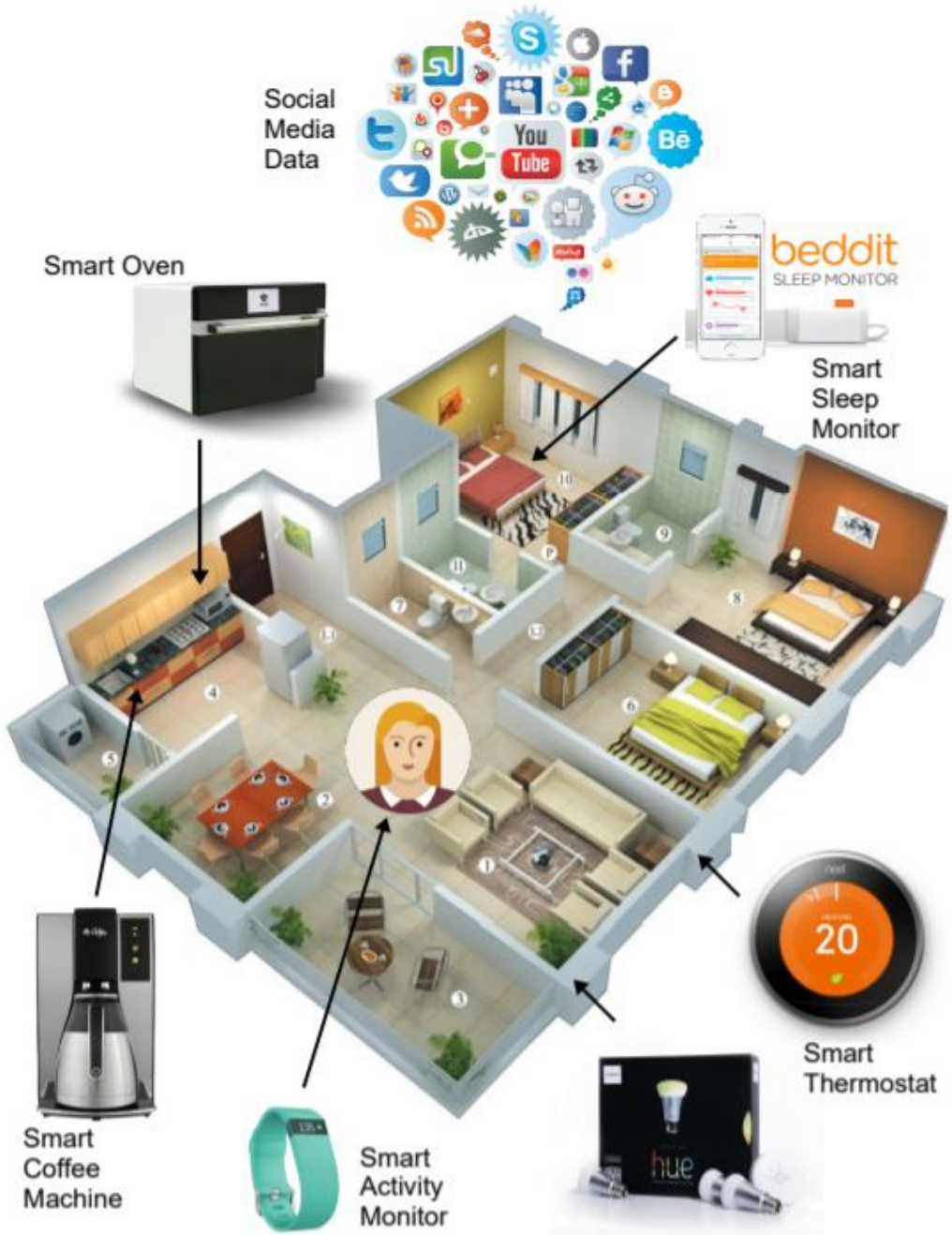
Smart Cities



Source: Sensing as a Service - Charith Perera , 2018

Smart Wearable /Smart Home





Applications

Telemetry
- Utility meters
- Parking meters
- Industrial meters
- Elevators
- Vending Machines

Fleet Management
- Cargo tracking
- Stock management
- Temperature control
- Route planning
- Order tracking
- Vehicle diagnostics

Service & Maintenance
- Industrial machines
- Vending machines
- ...

Telematics / Vehicle
- Navigation
- Traffic / weather info
- Road safety
- Vehicle diagnostics
- Location services



Security & Surveillance
- Public surveillance
- Asset monitoring
- Congestion and movement monitoring
- Urban management

Home Applications
- Heating control
- Electrical appliances
- Alarms and security
- Surveillance cameras
- Garage & garden

e-Health Applications
- Patient monitoring
- Remote diagnostics
- Activity monitoring
- Lifestyle suggestions
- Personal security

Sales & Payment
- POS terminals
- Vending machines
- Gaming & entertainment

IoT at work: six real examples (2025-26)

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Utilities

Great Britain: 42 million smart meters - 72% of all meters

Agriculture

John Deere See & Spray: cameras spray only weeds - about 50% less herbicide

Logistics

Maersk: 4G/5G solar trackers on ~30% of its refrigerated containers

Industry

BlueScope steel: ~2,000 hours of downtime avoided by predictive maintenance

Healthcare

NHS England: ~12,600 'virtual ward' beds - patients monitored at home

Smart city

Prague: fill-level sensors cut glass-bin collections by 65%

IoT in defense and public safety

Ukraine 'Sky Fortress'

~9,500 phone-based acoustic sensors track drones - each under USD 500

Sensor fusion (NATO, 2025)

Exercise in Estonia fused Sky Fortress with radars into one picture

Border sensors

Poland and the Nordics buy mesh-networked ground sensors

Drones = mobile IoT

Sensor + radio + actuator; Ukraine built ~2.2 million in 2024

Predictive logistics

US Army: vehicle sensors forecast failures and spare parts

Networks that survive

Mesh radios keep links alive under jamming

Israel: IoT is already in the street

~70%

of Israel's water meters are smart (goal: 100% in 2026)

1 million+

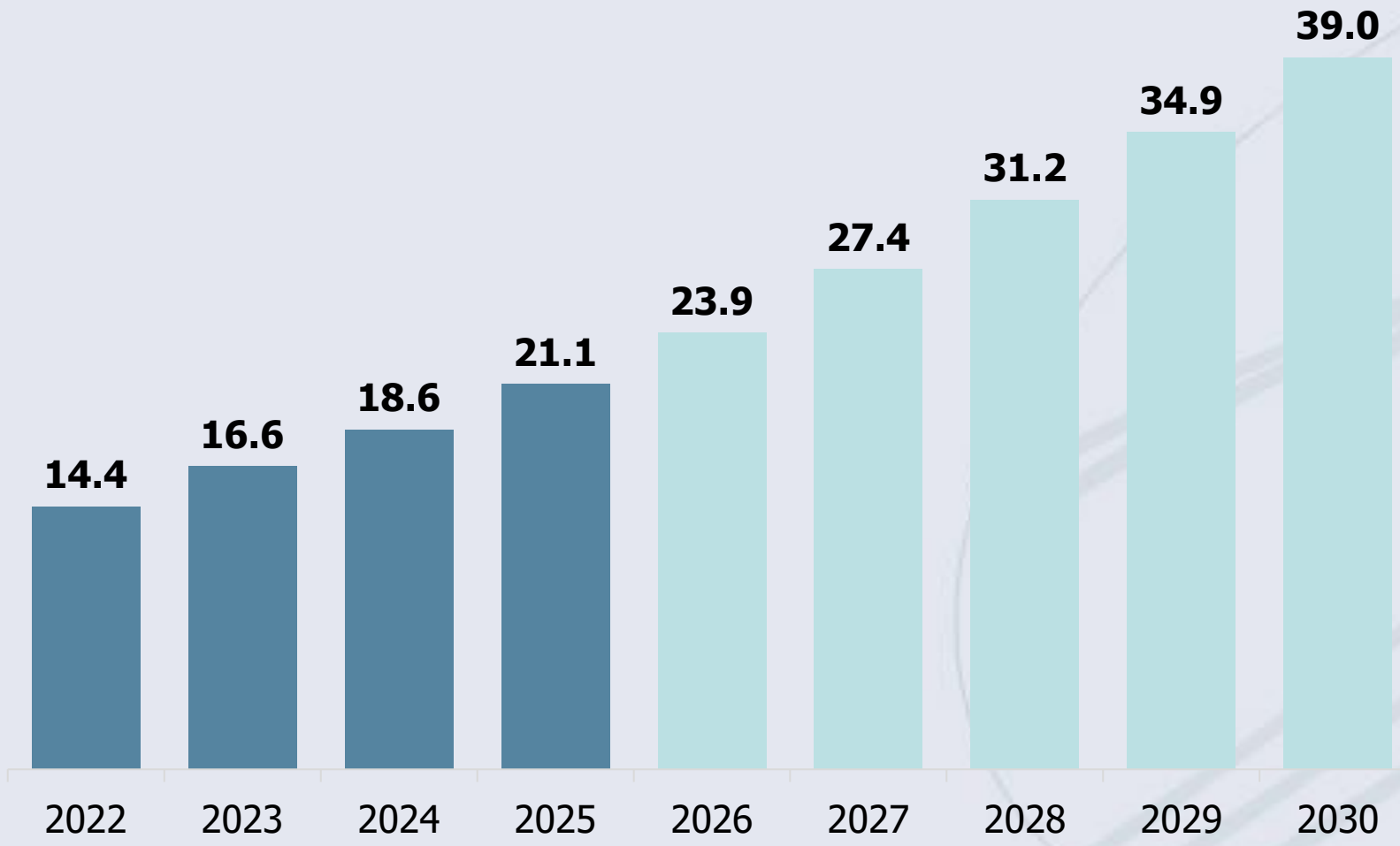
LoRa water meters and sensors in dozens of cities (Nov 2025)

2028

target: smart electricity meters in every home

- Smart water meters flagged leaks in 200,000+ homes in 2024
- The same LoRa networks also serve parking, street lighting, air quality
- Tomorrow: how such a city-wide LoRaWAN network actually works

Connected IoT devices worldwide (billions)



21.1 B

end of 2025

~2.6

devices per person

39 B

forecast for 2030

Source: IoT Analytics, State of IoT 2025 (Oct 2025). Light bars = forecast.

IoT by the numbers

32%

of IoT connections are Wi-Fi

24%

are Bluetooth

22%

are cellular (2G-5G, NB-IoT, LTE-M)

$324 B

enterprise IoT market, 2025 (+13%)

4.5 B

cellular IoT connections, end-2025

< 1%

of IoT devices have real on-device AI

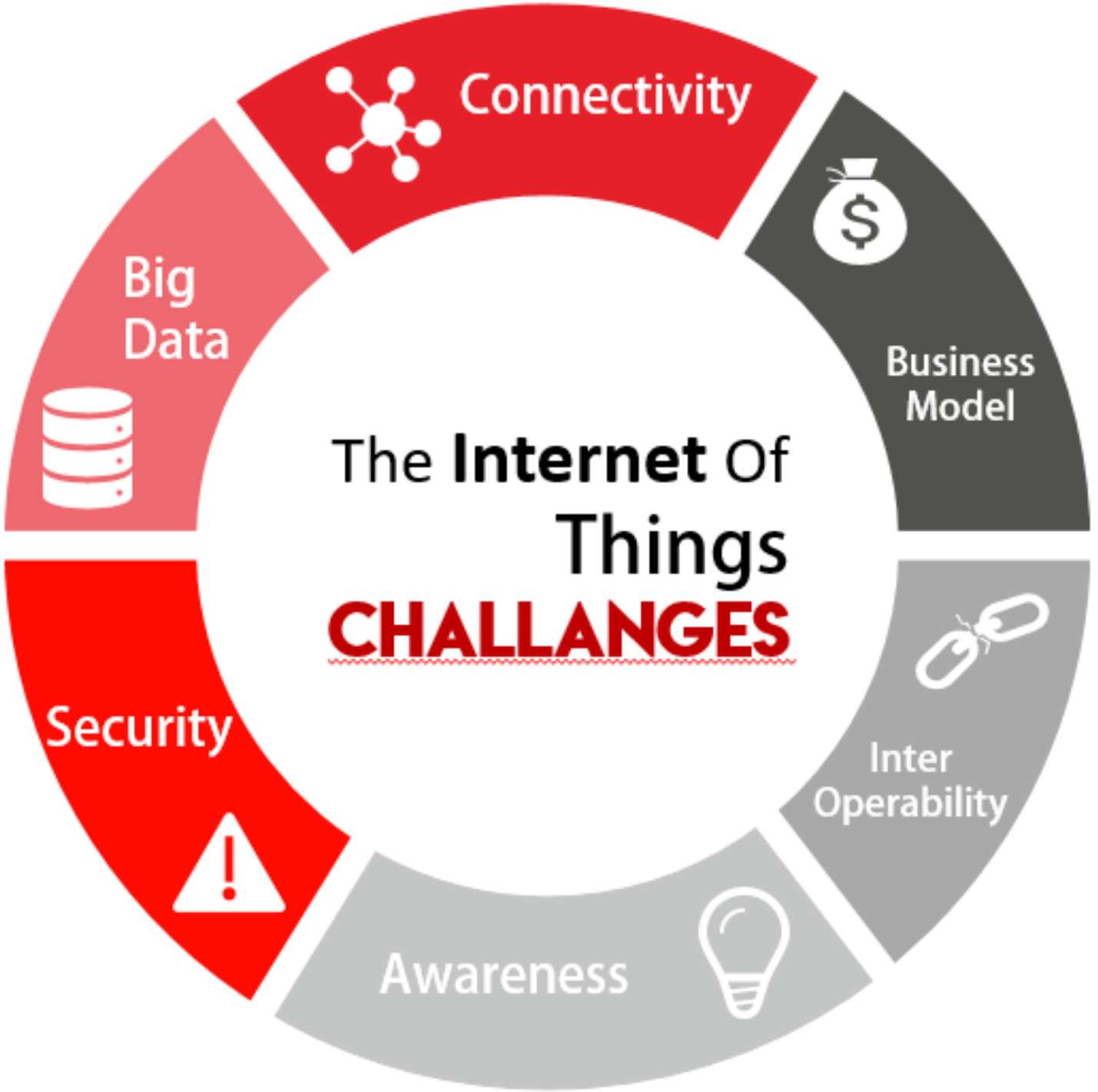
Differences Between H2H and M2M

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TOPICS	H2H	M2M / IoT
Entity #	There are a lot of us. But maybe not that much compared to M2M potential	M2M outnumbers human end users by order of magnitude
Data Volume	Most of the traffic is downloaded and requires significant amounts of bandwidth (file, Web and video streaming).	Traffic is mainly uploaded and most of it requires small amounts of bandwidth (video surveillance might require more).
Battery	We can buy a new one. We can recharge.	Auto-generating power or be self-sustaining for long periods.
Delay	We do have a tolerance for it even when it comes to voice	some applications are mainly for real-time control; urgent/emergency action would have little tolerance.
security	Worth comes to worth. Instinct tells us or we just know if it has been stolen.	Not so much here. Machine Communication Identity Module (MCIM) in Universal Integrated Circuit Card (UICC) or Trusted Environment (TRE).
Revenue	Good.	Low ARPD.
Value Chain	Well defined.	To be created/adapted, new opportunities on top of common infrastructure.
Reachability	satisfying.	Push/pull behavior. Might require longer dormant period, minimize signaling on control plane.

Characteristics (IoT v/s Cellular)

- **IoT communications are or should be:**
  - Low cost,
  - Low power,
  - Long battery duration,
  - High number of connections,
  - Different bitrate requirement,
  - Long range,
  - Low processing capacity,
  - Low storage capacity,
  - Small size devices,
  - Simple network architecture and protocols

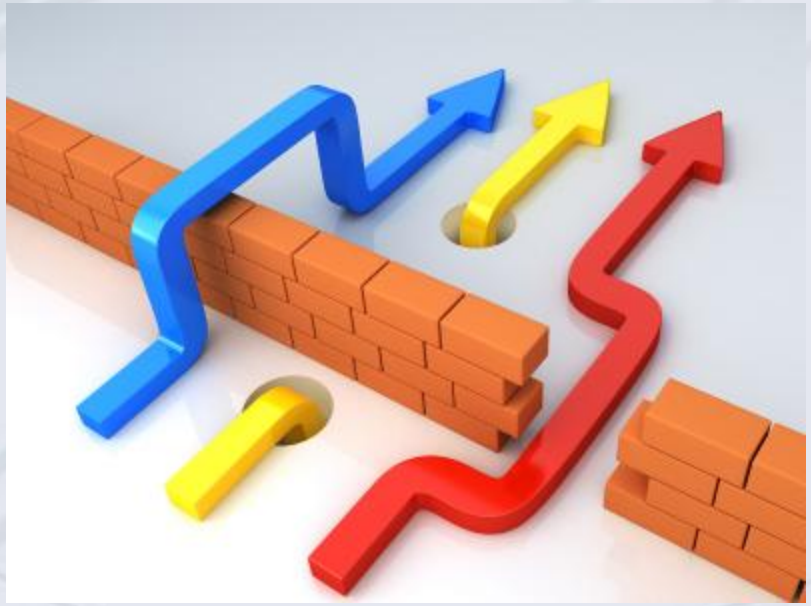

Connectivity
Business Model
Inter Operability
Awareness
Security
Big Data
The Internet Of Things
CHALLANGES

Challenges and Barriers to IoT

- Identification
- Deployment of IPv6
- Sensor energy
- Standards



- Bandwidth management
- Security
- AAA



When IoT becomes a weapon (2024-2026)

31.4 Tbps

record DDoS (Q4 2025) - a botnet of mostly Android TV boxes

130 -> 805

attacks above 1 Tbps Q1 -> Q2 2026

millions

devices sold pre-infected (FBI, BADBOX 2.0, 2025)

Poland, Dec 2025	30+ wind and solar sites attacked; controllers damaged, no blackout
Norway, Apr 2025	hackers opened a dam valve for about 4 hours via a remote panel
USA, Sep 2024	FBI takedown of a 260,000-device camera and router botnet

Israel: cameras turned into enemy sensors

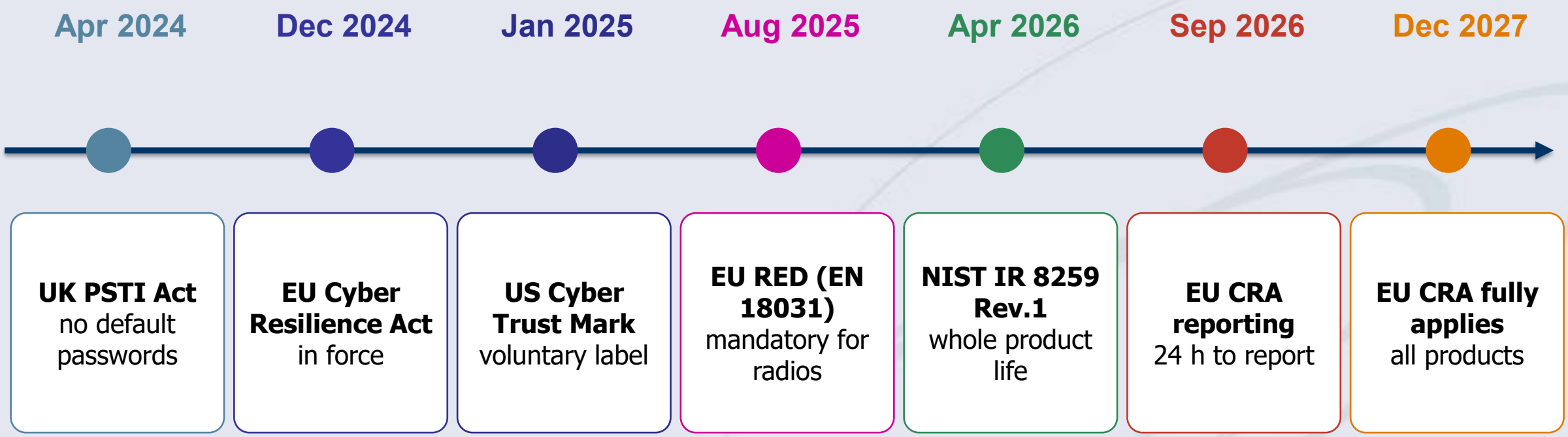


- June 2025: Iranian operators tried to watch where missiles landed - through private cameras
- INCD: a street camera overlooking the Weizmann Institute was hijacked before the strike
- 2026 war: INCD reported dozens of camera breaches (March 2026)

INCD advice - the same rules as the new regulations

No direct Internet access	**Unique strong password**	**Update the firmware**	**Two-factor login**

Regulation has arrived: IoT security timeline



What they all ask for:

- **no default passwords**
- **a way to report flaws**
- **security updates for a stated period**
- **secure by design**

Who writes the IoT standards?

Architecture and vocabulary	ISO/IEC JTC 1/SC 41 (ISO/IEC 30141:2024), ITU-T (Y.4000)
Applications and messaging	OASIS (MQTT), CSA (Matter, Zigbee), OMA (LwM2M), Eclipse (Sparkplug)
Internet protocols	IETF: IPv6, 6LoWPAN, CoAP, TLS
Radios and cellular	IEEE (Wi-Fi, 802.15.4), Bluetooth SIG, LoRa Alliance, 3GPP (NB-IoT, LTE-M, 5G)
Security baselines	ETSI EN 303 645, NIST IR 8259, IEC 62443, EN 18031

Many bodies, many standards - the price of a huge, diverse market.

DATA NETWORKING

Why layers? Think of sending a letter

The post office		The network	
 The letter	⟷	Application data	temp = 23.5
 Name on the envelope	⟷	Transport: port	MQTT, port 8883
 Street address	⟷	Network: IP address	18.192.x.x
 Mail-bag, next truck	⟷	Link: Wi-Fi frame	next hop only
 Road or air	⟷	Physical: radio, cable	2.4 GHz

Each layer adds its own envelope - the receiver opens them in reverse order.

OSI 7 Layer Model

- OSI - Open Systems Interconnection (Basic Reference Model)
- Each level is an independent set of protocols
- Each level can be change seamlessly

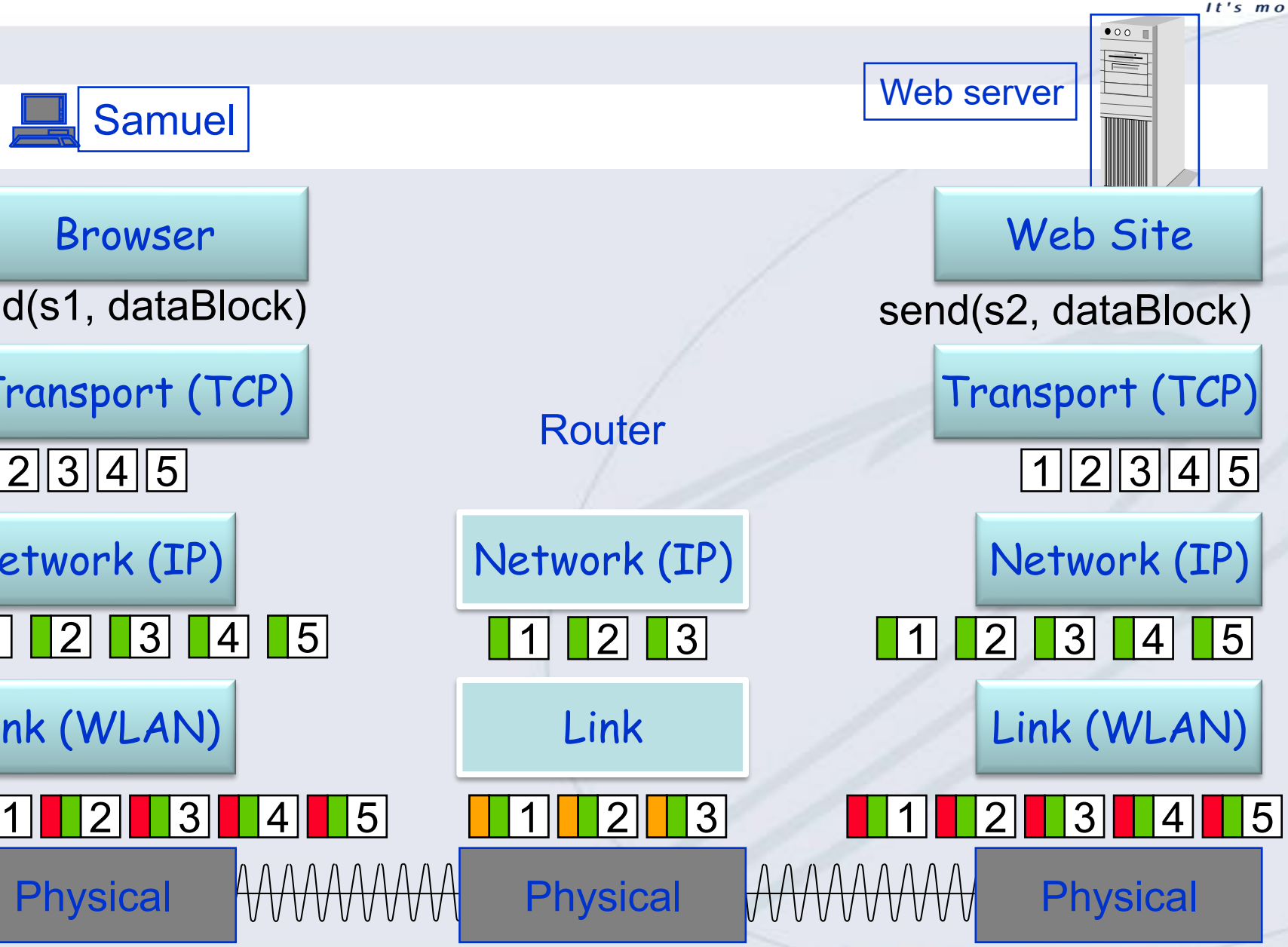


The 7 OSI layers in plain words

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#	Layer	Its job	IoT example
7	Application	The service the device wants	MQTT, HTTP, CoAP
6	Presentation	Format, compress, encrypt	JSON, CBOR, TLS
5	Session	Open, keep and close a dialogue	MQTT session, keep-alive
4	Transport	Right application; reliable or fast	TCP, UDP, ports
3	Network	Addresses and routes	IPv4, IPv6
2	Data link	One hop; who may talk now	Wi-Fi, BLE, LoRaWAN MAC
1	Physical	Bits as signals	radio, cable, fiber

Our focus: layers 7, 5 and 4 - and just enough of 1-3 to follow the data.

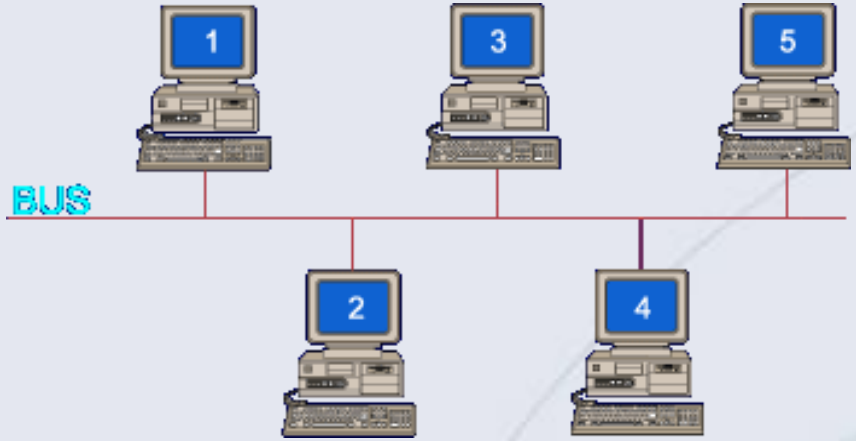

Samuel
Web server
Browser
Web Site
read(s1, dataBlock)
send(s2, dataBlock)
Transport (TCP)
Transport (TCP)
Router
1 2 3 4 5
1 2 3 4 5
Network (IP)
Network (IP)
Network (IP)
1 2 3 4 5
1 2 3
1 2 3 4 5
Link (WLAN)
Link
Link (WLAN)
1 2 3 4 5
1 2 3
1 2 3 4 5
Physical
Physical
Physical

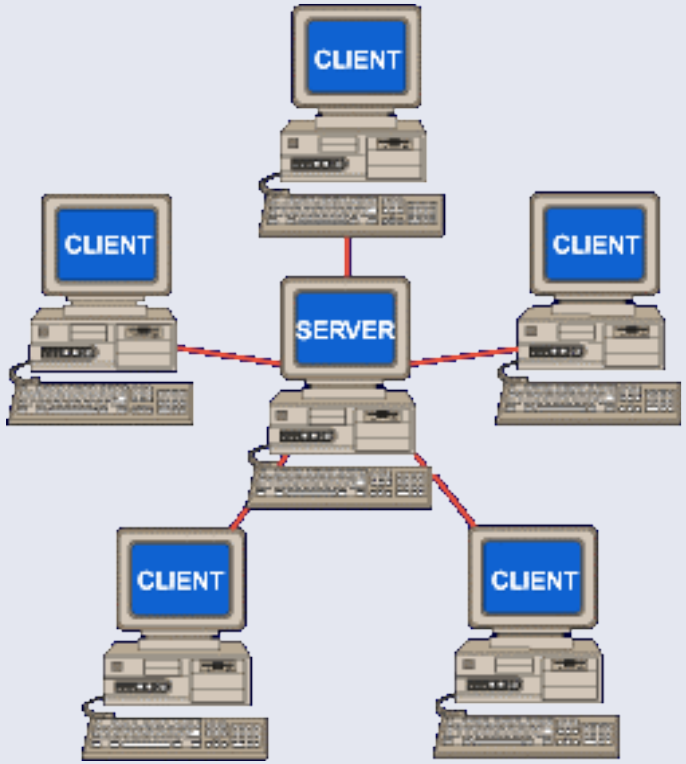
Where IoT protocols live

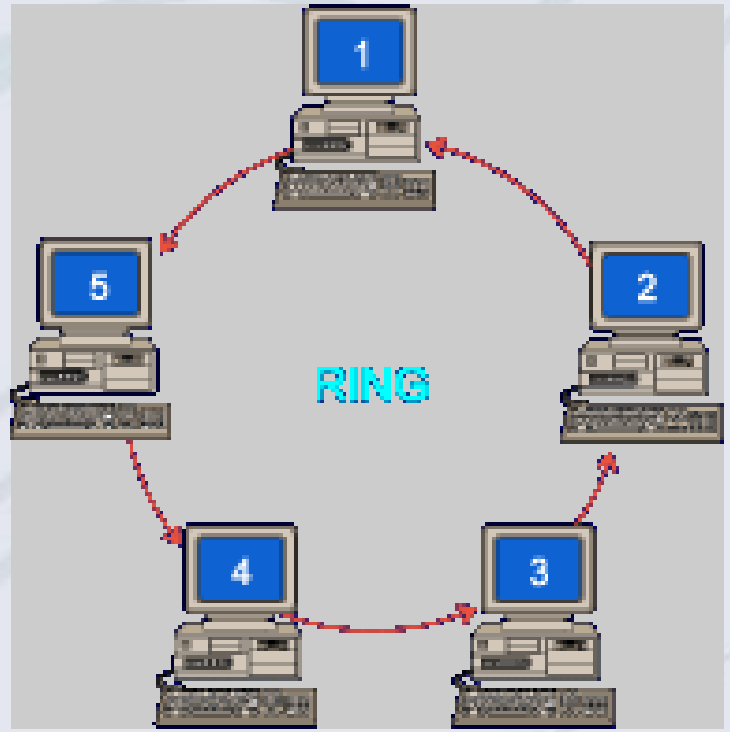
	Classic Internet	IP-based IoT	Non-IP IoT
Application	HTTP	MQTT, CoAP	raw bytes
Security	TLS	TLS, DTLS	link encryption
Transport	TCP	TCP or UDP	-
Network	IPv4 / IPv6	IPv6, 6LoWPAN	no IP
Link / radio	Ethernet, Wi-Fi	Thread, NB-IoT	LoRaWAN, BLE, Zigbee

Many IoT devices never speak IP - a gateway translates them into the Internet.

NETWORK TOPOLOGIES (shape)

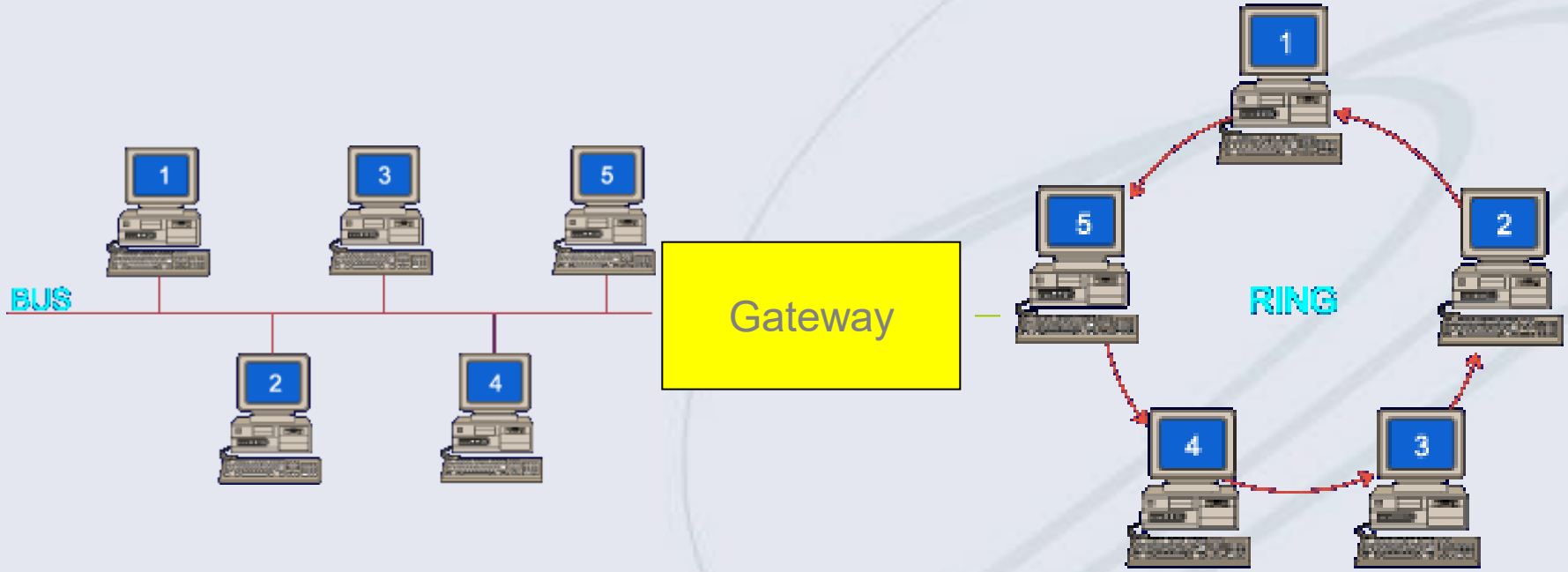






Gateway

Often used to connect a LAN with a WAN.
Gateways join two or more different networks together.

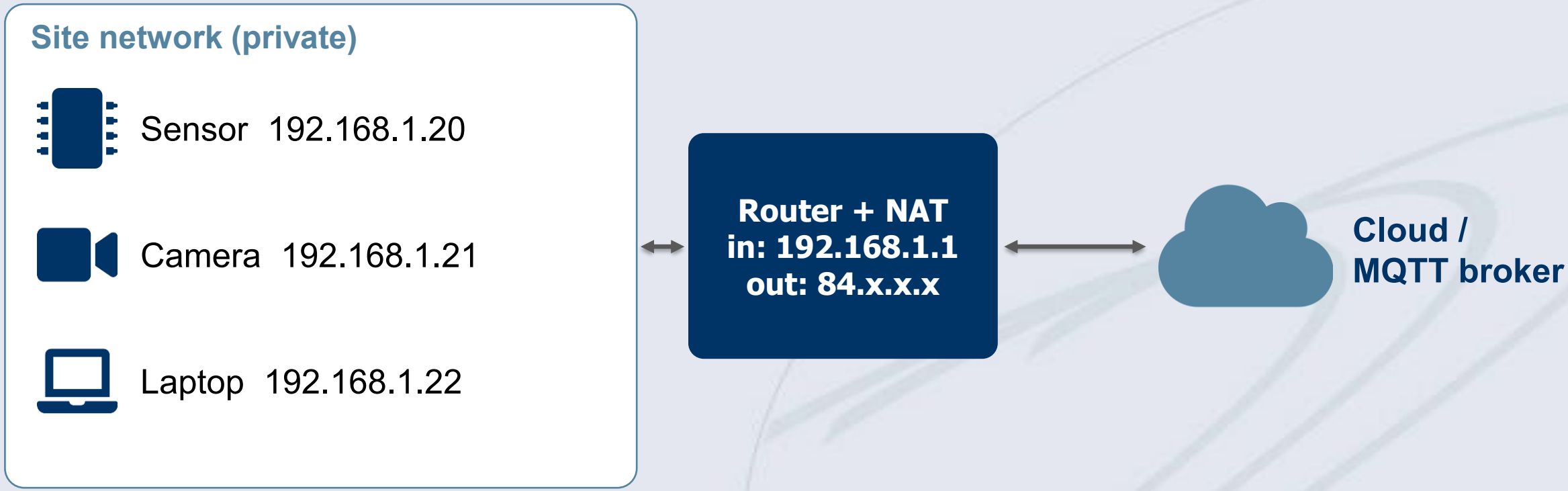


IP Address

- Unique addresses in the world
- An IP address is 32 bits, noted in dotted decimal notation:
  192.78.32.2

- Host and Prefix Part
  - An IP address has a prefix and a host part:
    - **prefix:host**
  - Prefix identifies a subnetwork
    - used for locating a subnetwork - routing
  - Prefix is usually identified in a host using a "subnet mask"

Private addresses and NAT



- Private addresses are not reachable from the Internet
- So the device dials OUT and keeps the connection open - commands come back on it

IPv6 and IoT

4.3 billion

IPv4 addresses in total

21+ billion

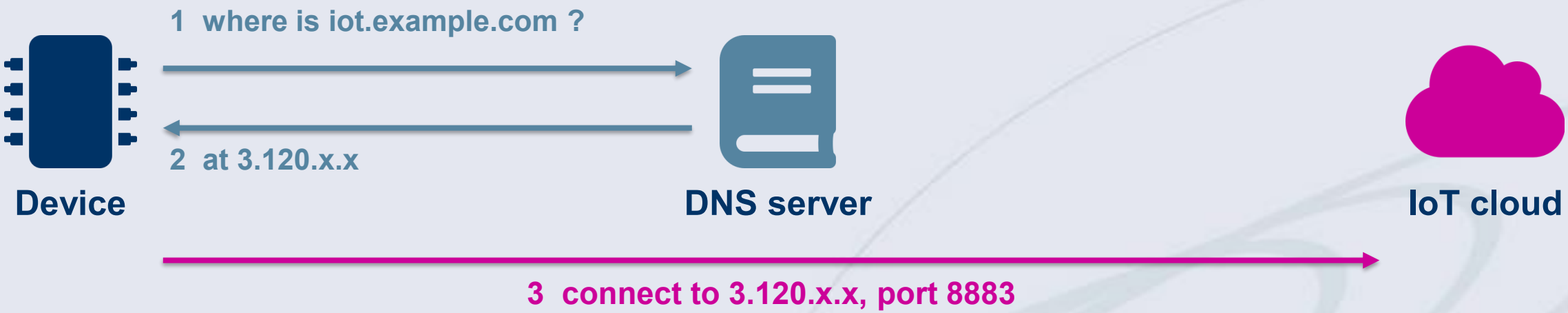
connected IoT devices (2025)

3.4 x 10^38

IPv6 addresses

- IPv4 survives thanks to NAT - fine for devices that only dial out
- IPv6: every device can have its own global address
- **6LoWPAN** squeezes IPv6 into tiny radio frames; **Thread** (used by Matter) is IPv6 end to end
- Cellular IoT and the big clouds run both (dual-stack)

DNS: the Internet's phone book



- Names for people and devices, numbers for the network - DNS translates
- The first thing a device does on a new network - and a classic failure point
- Never hard-code a cloud IP address in a device: names stay, addresses change
- Wireshark filter: **dns**

DHCP

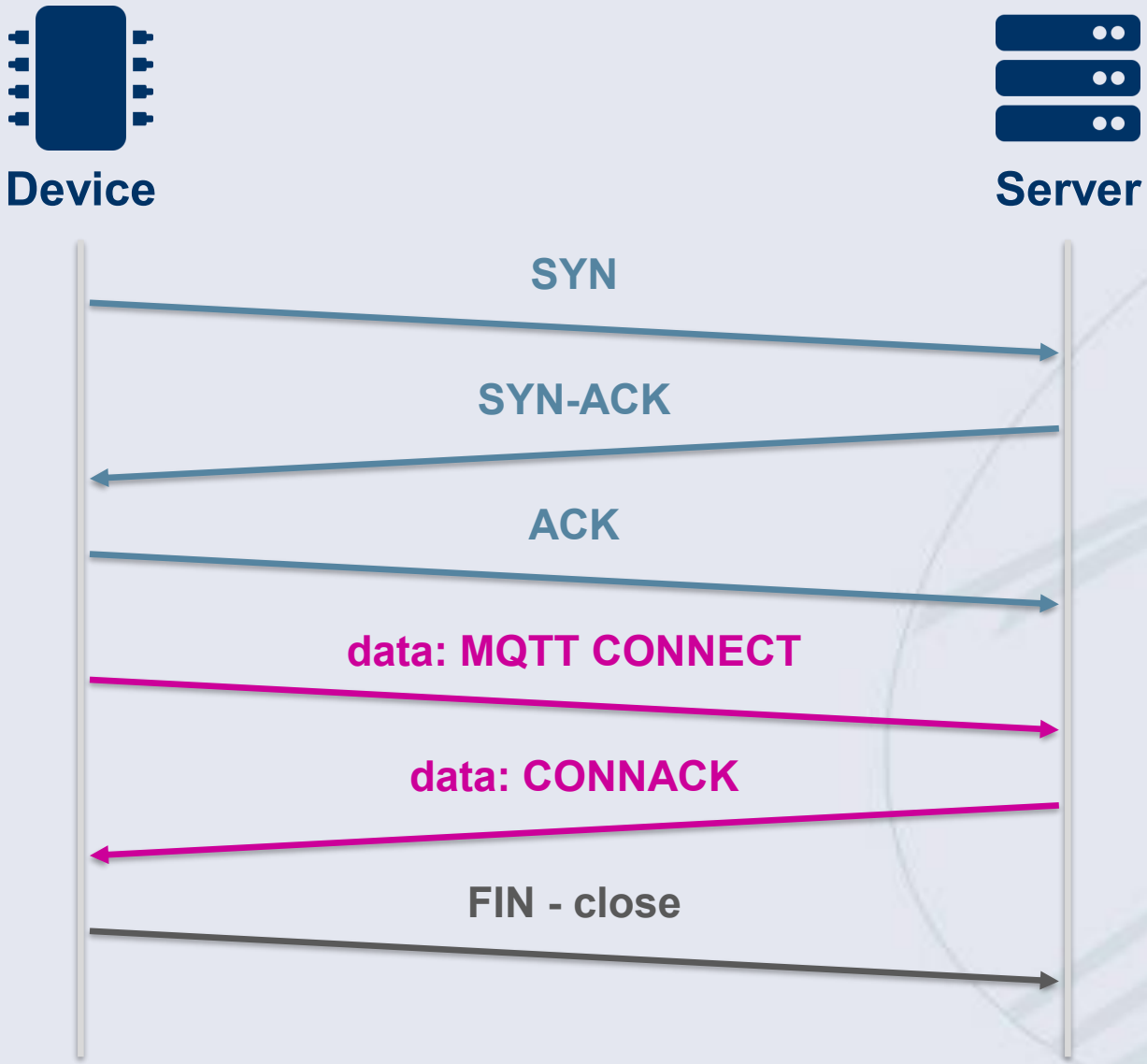
- Dynamic Host Configuration Protocol (DHCP) is designed, to dynamically configure TCP/IP hosts in a centralized manner from DHCP server.
- DHCP server maintains a collection of configuration parameters, such as IP addresses, subnet mask, default gateway IP address, to make a configured host work in the network.
- A DHCP client queries the server for the configuration parameters.
- The DHCP server returns configuration parameters to the client.

Layer 4: ports - the apartment number

Port	Protocol	What IoT uses it for
53	DNS (UDP)	find the server's address
80 / 443	HTTP / HTTPS (TCP)	web and cloud APIs
1883	MQTT (TCP)	message broker, no encryption
8883	MQTT over TLS (TCP)	secure MQTT - AWS IoT Core
5683	CoAP (UDP)	tiny battery devices
1700	LoRa packet forwarder (UDP)	LoRaWAN gateway to network server

IP address = which building. Port = which apartment (which program) inside it.

TCP: a handshake before talking



- 3 messages before any data
- Every byte is numbered and acknowledged - losses are re-sent
- Costly on a battery: MQTT keeps ONE connection open for hours
- Wireshark: **tcp.flags.syn == 1**

TCP or UDP? Reliable vs fast

	TCP	UDP
Connection	handshake first	none - just send
Reliability	acknowledged, re-sent, in order	best effort
Overhead	20+ byte header, ACKs	8 byte header
IoT examples	MQTT, HTTPS, AWS IoT	DNS, CoAP, LoRa forwarder
Analogy	registered mail	a postcard

Modern twist: QUIC (the base of HTTP/3) adds reliability on top of UDP.

Layer 5 in IoT: sessions that stay alive



- A session remembers who you are and what you subscribed to
- **Keep-alive**: small pings, so both sides notice a dead link
- **Last Will**: the broker announces "site-7 offline" for you
- Reconnect and receive what you missed - all of this is MQTT, Day 2

Layer 6: one reading, three formats

Format	What is sent	Size
JSON (text)	{"temp":23.5,"bat":87}	22 bytes
CBOR (binary JSON)	A2 64 74 65 6D 70 F9 4D E0 ...	15 bytes
Packed binary	00 EB 57	3 bytes

- JSON is easy to read - but every character costs
- A long-range LoRaWAN message may carry only about 50 bytes
- So devices send compact bytes; a **decoder** turns them back into JSON (Day 2)

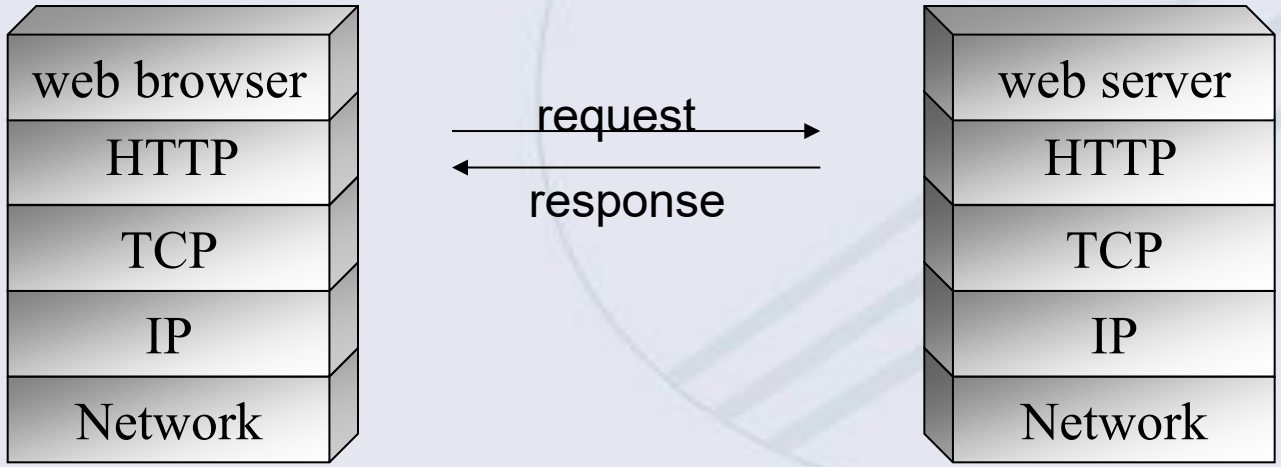
TLS: the padlock



- **Privacy** (encryption), **integrity** (no tampering), **identity** (the real server)
- HTTPS = HTTP + TLS; MQTT + TLS = port 8883
- Normal TLS proves only the server **- mutual TLS** (Day 3) makes the device prove itself too

HyperText Transfer Protocol (HTTP)

- Application layer protocol
- Distributes information in the WWW
- Based on the client/server architecture
  - HTTP client (web browser): sends a request to a server for a file
  - HTTP server (web server): well-known port number 80, responds with the requested file if it is available
  - A single TCP connection is used

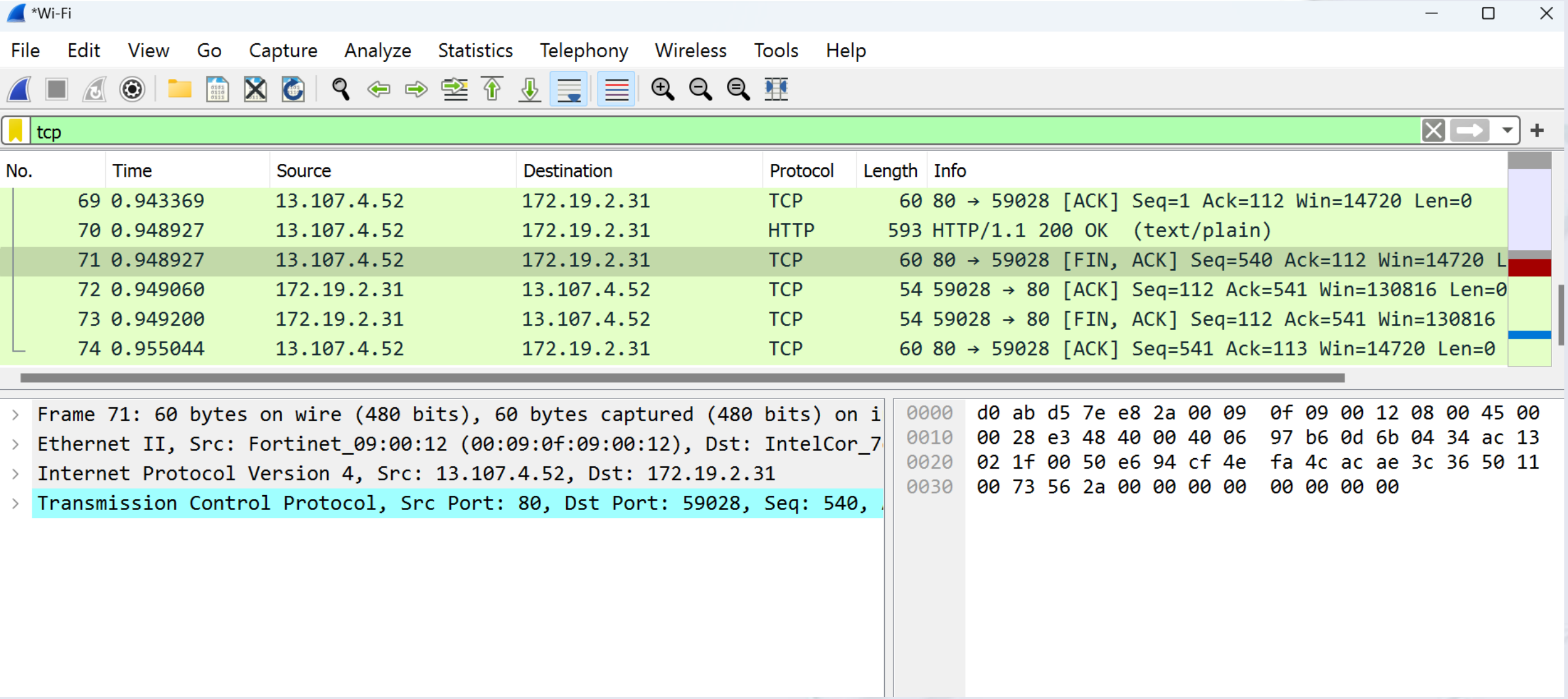


Layer 7: IoT application protocols

Protocol	Pattern	Best for
HTTP / REST	request - response	cloud APIs, occasional uploads
MQTT	publish - subscribe (broker)	telemetry and commands
CoAP	request - response over UDP	tiny battery devices
LwM2M	device management	firmware, configuration
AMQP	message queues	enterprise back-ends

HTTP: the device asks, the server answers.
MQTT: everyone talks to a broker - the star of Day 2.

Wireshark: every layer of every packet



Wireshark: cheat sheet and a 5-minute try

Display filter	Shows
dns	name look-ups
tcp.flags.syn == 1	new connections
tls.handshake	TLS handshakes
mqtt	MQTT (Day 2-3)
ip.addr == 192.168.1.20	one device

Three panes: packet list, packet details (the layers), raw bytes.

Try it - 5 minutes

- Capture on Wi-Fi
- Open a web site
- Filter **dns** - find question and answer
- Filter **tcp.flags.syn == 1**
- Name the layers of one packet

Capture only where you are authorized.

HARDWARE INTERFACES

How sensors talk to the controller: GPIO, UART, I2C, SPI

Inside an IoT device



Sensors

temperature, motion, light



Radio

BLE, Wi-Fi, LoRa, cellular

Microcontroller (MCU)



Actuators

LED, relay, motor, display



Power

battery, harvesting

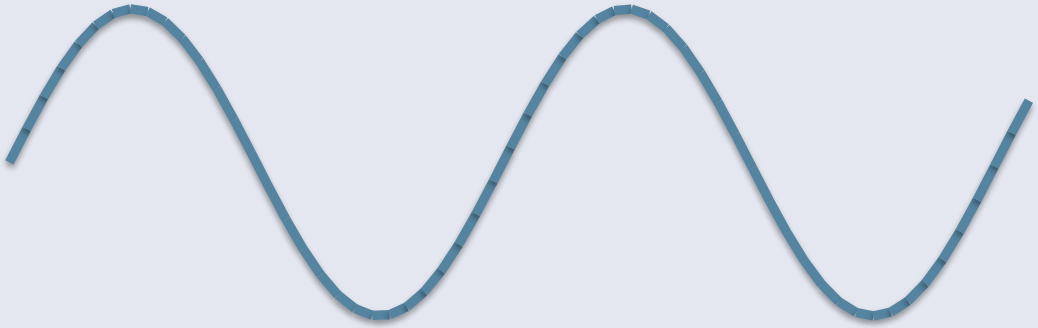
Today's question: what are the wires between these chips?

A microcontroller is not a small PC

	micro:bit V2	Raspberry Pi 5	Laptop
Processor	64 MHz, 1 core	2.4 GHz, 4 cores	3-5 GHz, many cores
Memory	128 KB RAM	2-16 GB RAM	16-64 GB RAM
Software	one program, no OS	Linux	Windows / macOS
Power	milliwatts	watts	tens of watts
IoT role	sensor node	gateway, edge	management

Kilobytes of memory and milliwatts of power: that is why IoT protocols must be small, simple and sleepy.

Analog vs digital - and the ADC



Analog: any value, continuously



Digital: only 0 or 1



- The ADC (analog-to-digital converter) turns a voltage into a number
- micro:bit: 0-3.3 V reads as 0-1023 - half the voltage is about 512

GPIO: one pin, one bit

LOGTEL
It's more than knowledge

Input

- button pressed?
- door open?
- motion?

Output

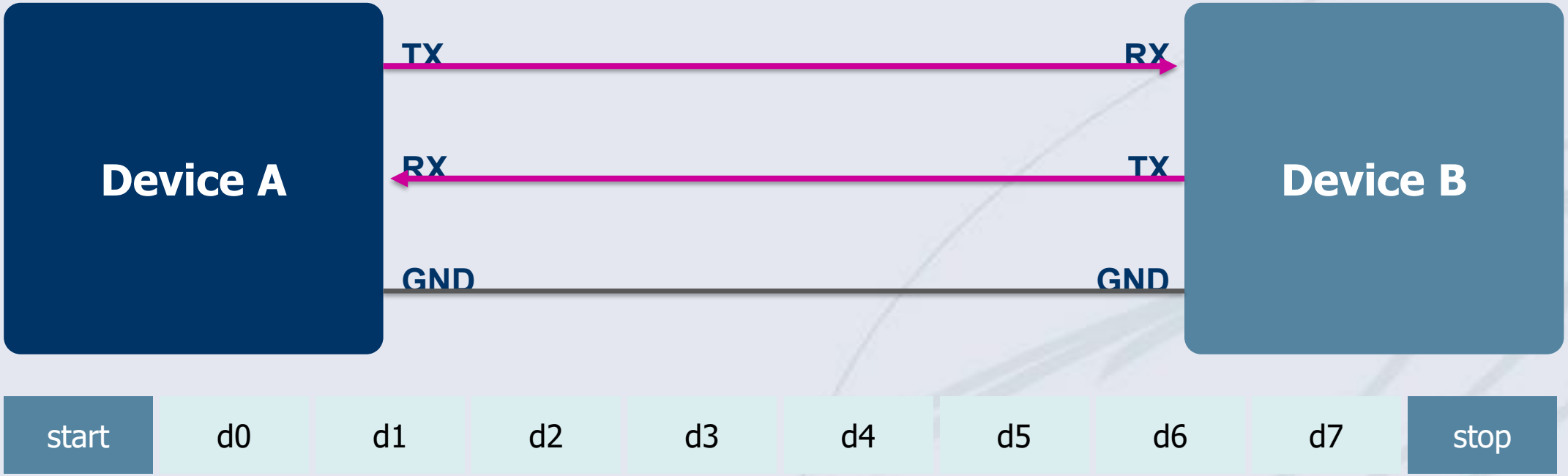
- LED on / off
- buzzer
- relay: switch a pump

PWM / analog

- dim an LED
- drive a servo
- read a light sensor

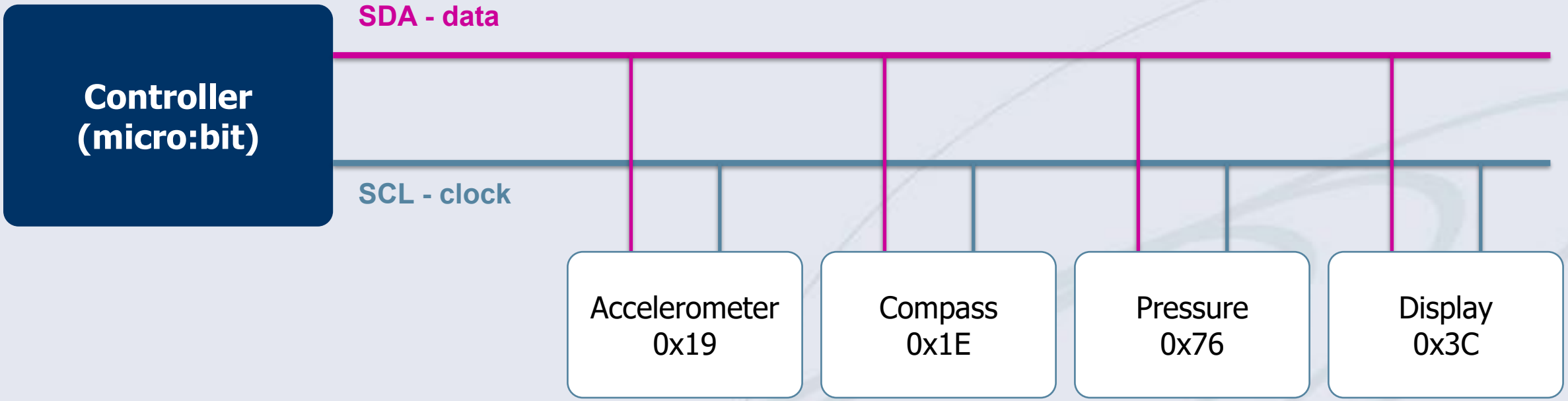
- **General Purpose Input/Output**: a pin read or driven high (3.3 V) or low (0 V)
- No protocol, no address - one signal per pin
- micro:bit: big rings 0, 1, 2 - and buttons A/B are GPIO inputs

UART: two wires and an agreed speed



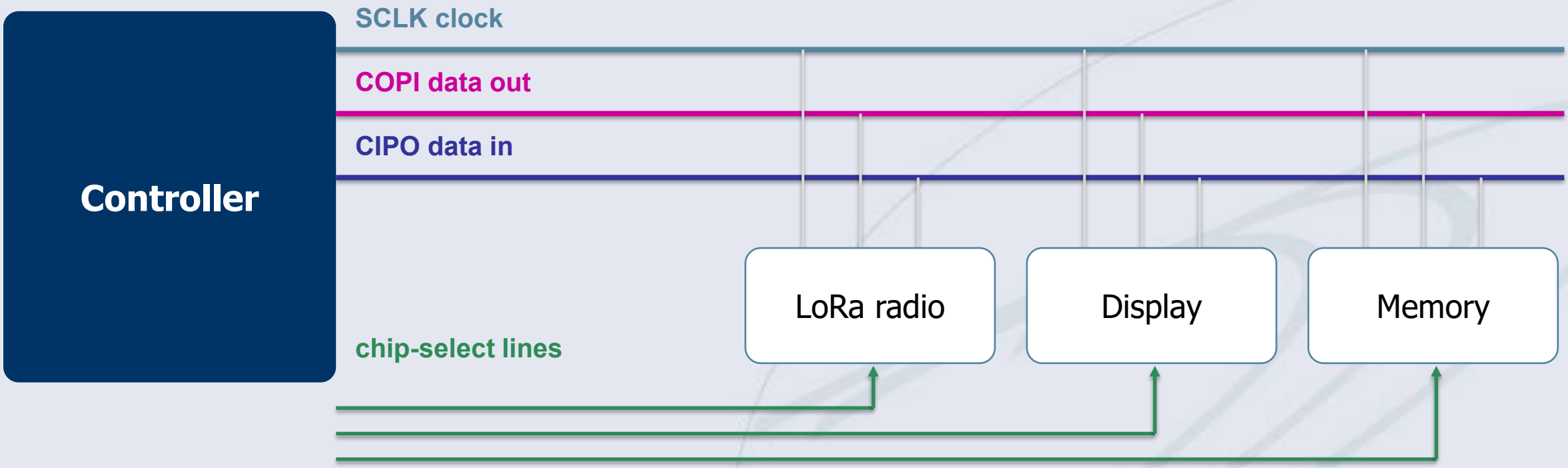
- No clock wire - both sides agree on the speed (baud), e.g. 115200
- GPS modules, modems (AT commands), debug consoles
- Our lab: the micro:bit drives its Wi-Fi chip over UART (AT commands)

I2C: two wires, many addressed devices



- The controller calls each device by its **address**
- 100 kbit/s to 3.4 Mbit/s; short, on-board wiring
- micro:bit V2: the motion sensor sits on an internal I2C bus

SPI: faster, one 'chip select' per device



- 4 wires + one chip select per device; MHz to tens of MHz
- Radios, displays, memory - every LoRa module is an SPI device

Choosing an interface

	Wires	Speed	Devices	Typical use
GPIO	1 per signal	-	1	buttons, LEDs, relays
UART	2 + ground	up to ~1 Mbit/s	2	GPS, modems, PC
I2C	2 + ground	up to 3.4 Mbit/s	many	sensors
SPI	4 + 1 each	tens of Mbit/s	several	radios, displays

RS-485 / Modbus

long cables in factories and utilities

CAN bus

vehicles, drones, robots

MIPI I3C

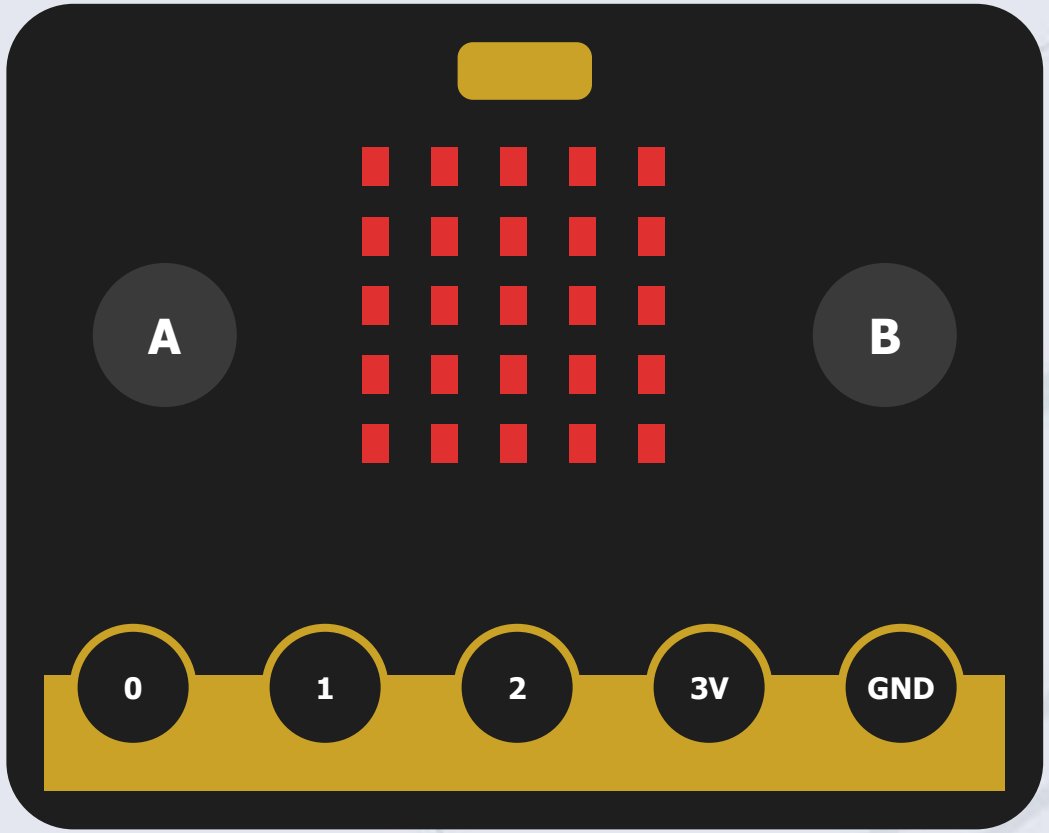
faster, lower-power successor of I2C

Our lab controller: BBC micro:bit V2

64 MHz processor
128 KB RAM

Bluetooth LE +
2.4 GHz radio

USB: power, flash,
serial to PC

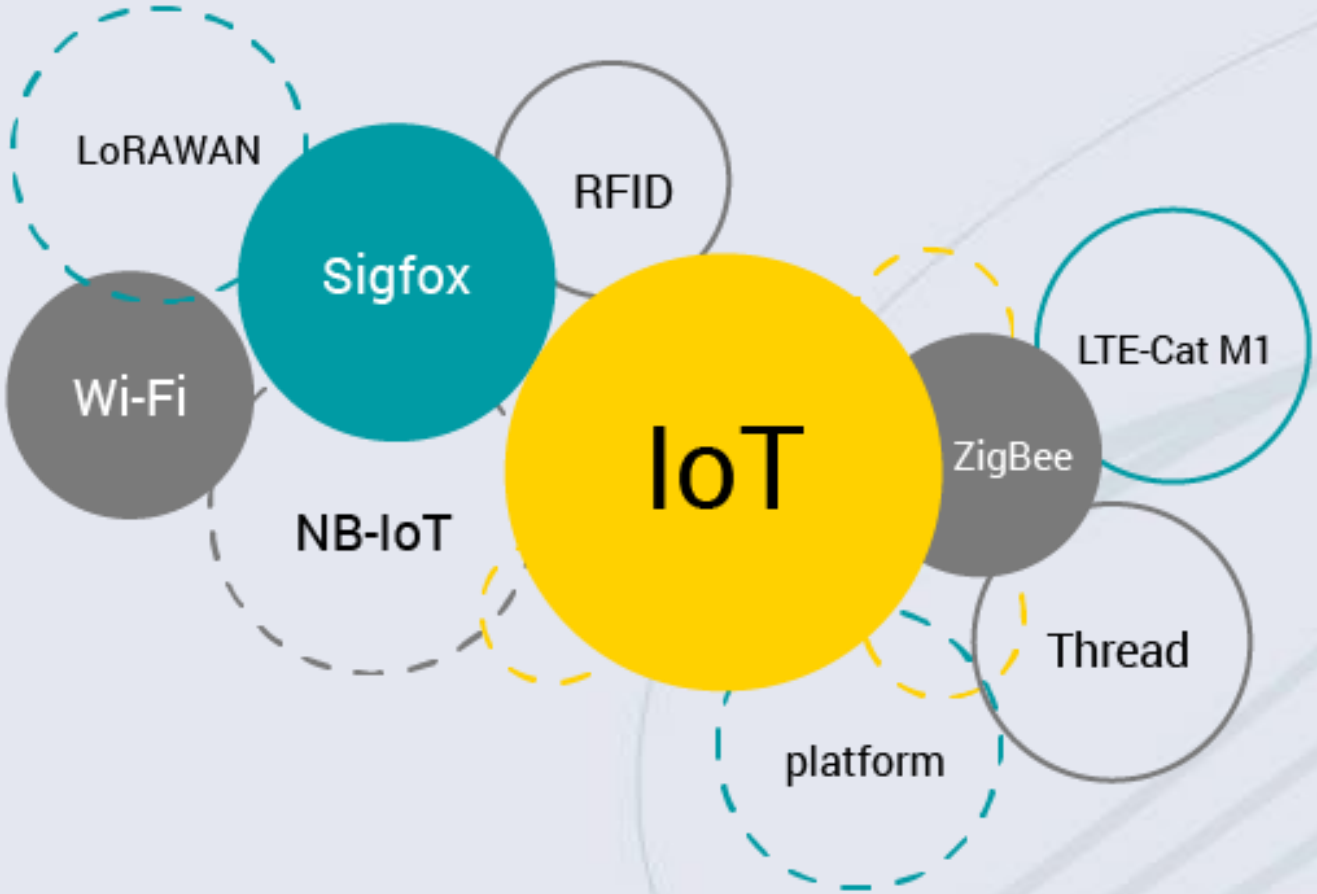


Motion sensor
(on I2C)

Temperature, light,
mic, speaker

Edge pins: GPIO,
I2C, SPI, UART

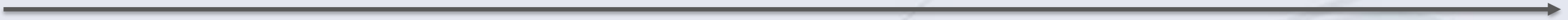
In the lab it sits on an ELECFREAKS IoT:bit board that adds Wi-Fi - the way a tiny sensor node reaches a network.



IoT technologies

Radio basics for non-engineers

13.56 MHz	sub-GHz	2.4 GHz	5 / 6 GHz	6-8 GHz
NFC	LoRa, HaLow, RFID	Wi-Fi, Bluetooth, Zigbee	Wi-Fi 5-7	UWB



lower frequency — higher frequency

Lower frequency

- travels farther
- passes walls
- less data

Higher frequency

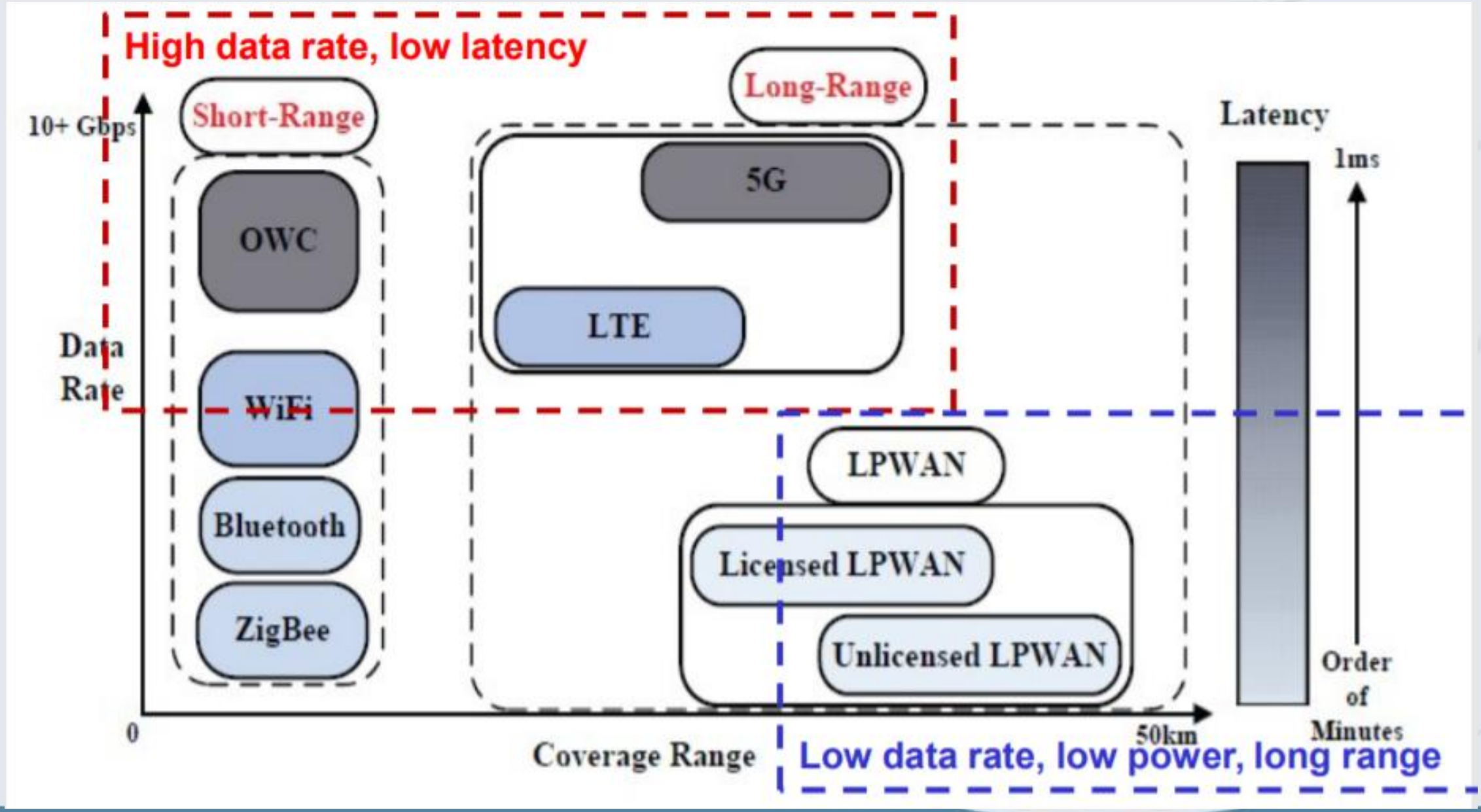
- shorter range
- easily blocked
- more data

Pick two

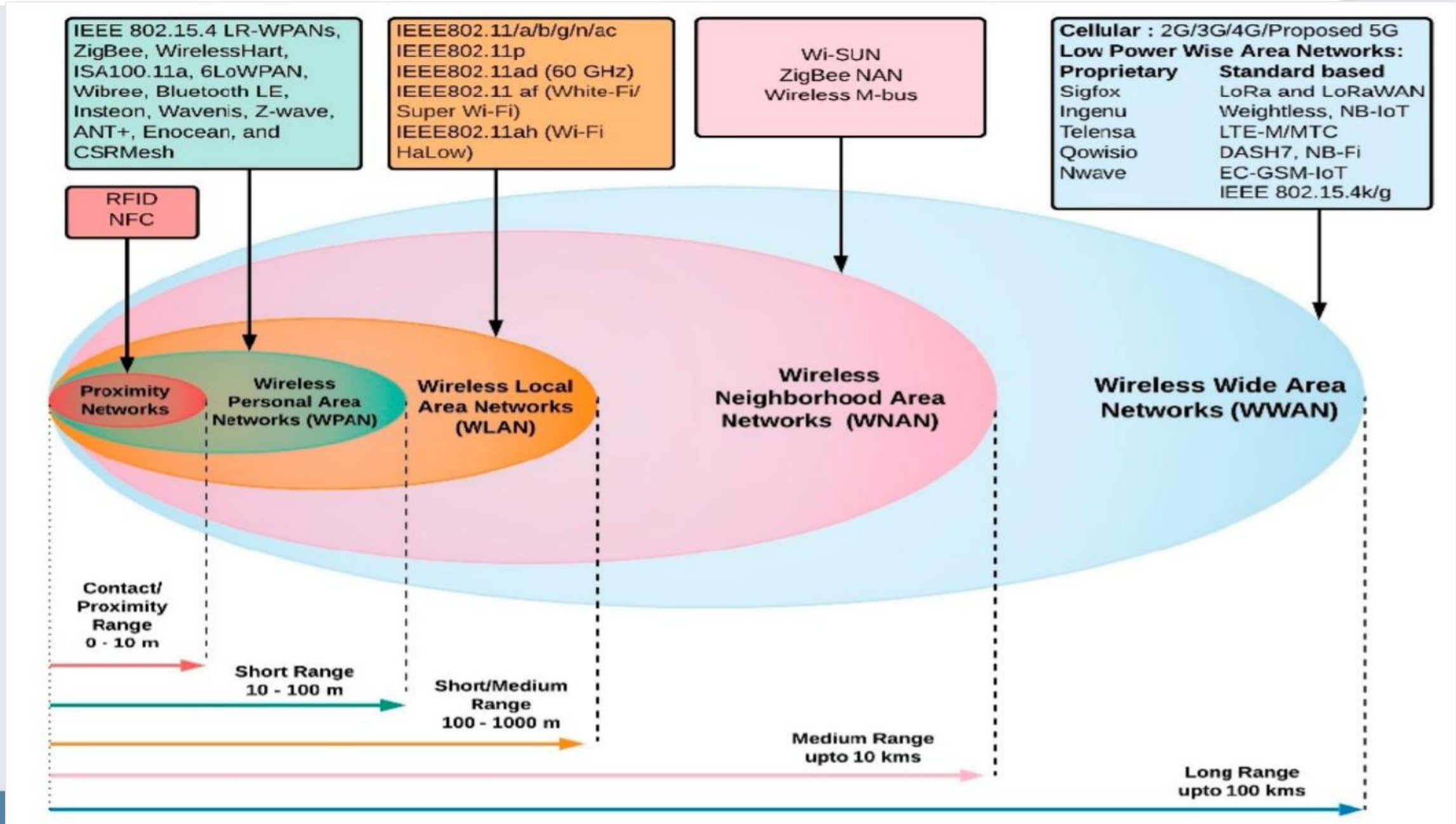
- long range
- high data rate
- low power

Licensed bands (cellular): paid and protected. Unlicensed (ISM): free for all - so crowded, with power and duty-cycle rules.

Short and Long Range



From NFC to 5G



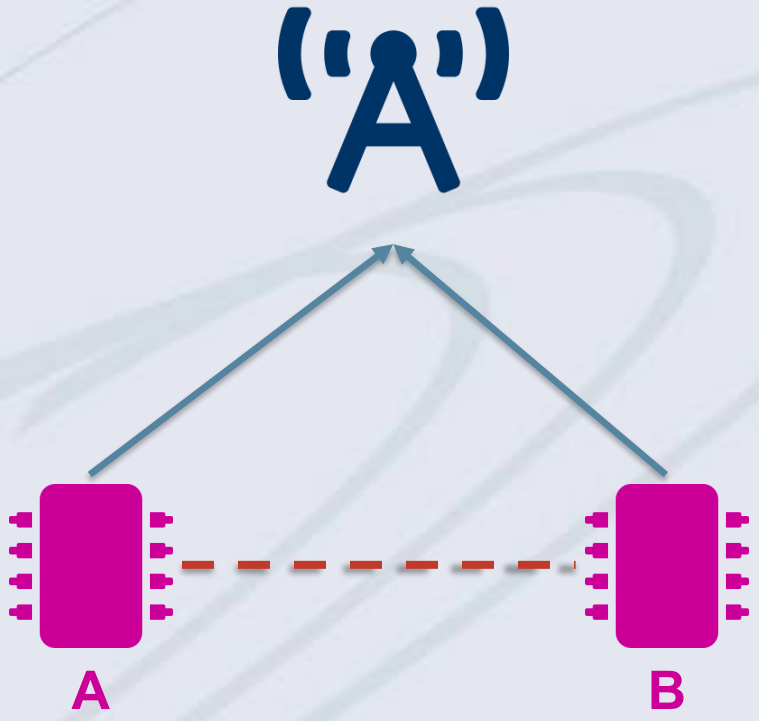
Short-range technologies at a glance

	Typical range	Data rate	Power	Typical IoT use
NFC	a few cm	up to 848 kbit/s	tag needs none	payment, access, pairing
RFID (UHF)	cm to several m	1,000 tags/s	tag needs none	inventory, logistics
Bluetooth LE	10-100 m	0.125-2 Mbit/s	very low	wearables, trackers, sensors
Zigbee / Thread	10-100 m + mesh	250 kbit/s	very low	home and building automation
Wi-Fi	20-150 m	up to Gbit/s	high	cameras, gateways
Wi-Fi HaLow	~1 km	up to tens of Mbit/s	low-medium	campus, farm, cameras
UWB	up to 100 m	up to ~27 Mbit/s	low	cm-accurate location

Typical values - real range depends on antennas, walls and settings.

Sharing the air

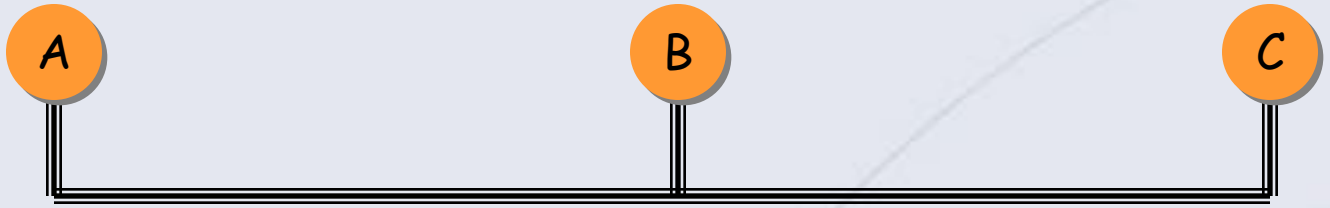
- On a cable you can listen while you talk - on radio you cannot
- Two devices may both reach the access point but not hear each other
- So Wi-Fi listens first, waits, and can reserve the air (RTS / CTS)
- More devices on a channel = more waiting
- LPWAN does the opposite: send rarely, accept collisions (Day 2)



the 'hidden node' problem

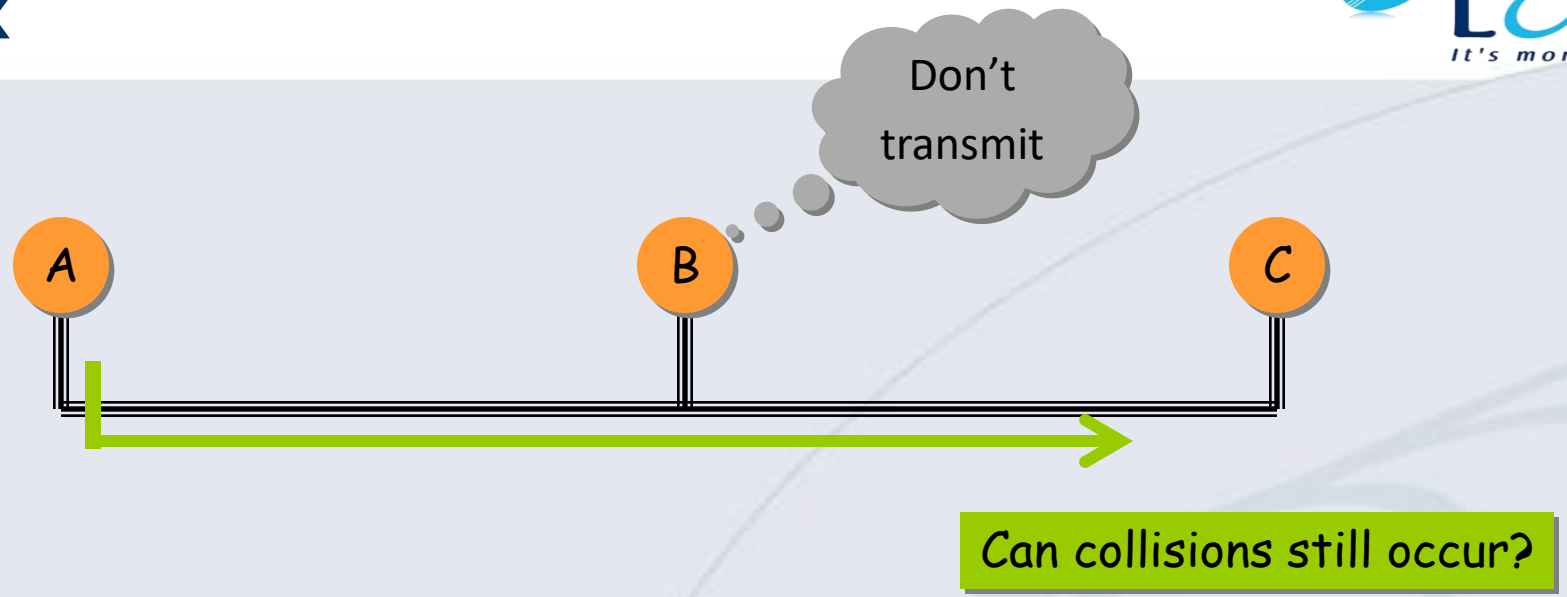
The Channel Access Problem

- Multiple nodes share a channel



- Pairwise communication desired
  - Simultaneous communication not possible
- MAC Protocols
  - Suggests a scheme to schedule communication
    - Maximize number of communications
    - Ensure fairness among all transmitters

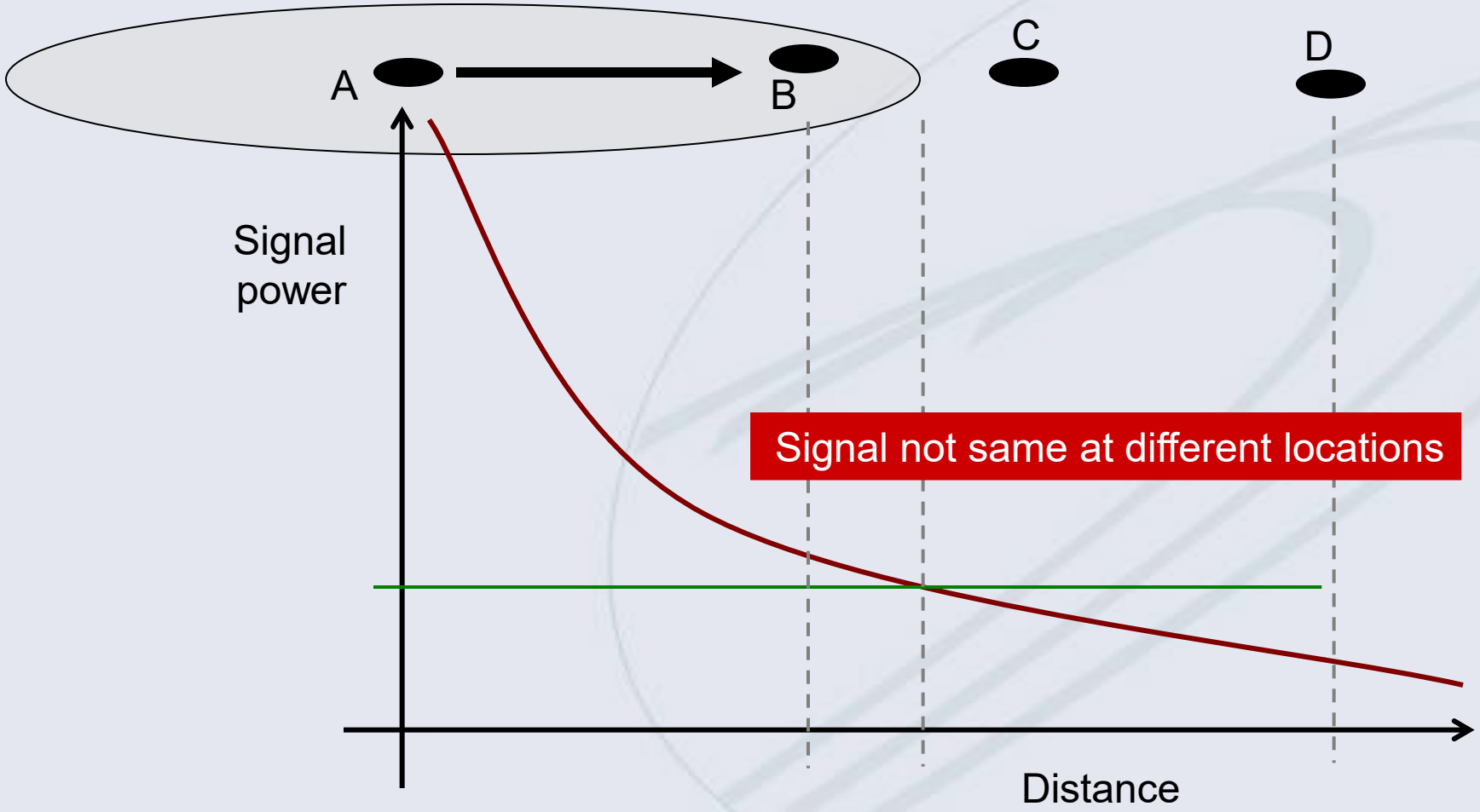
The Simple Fix



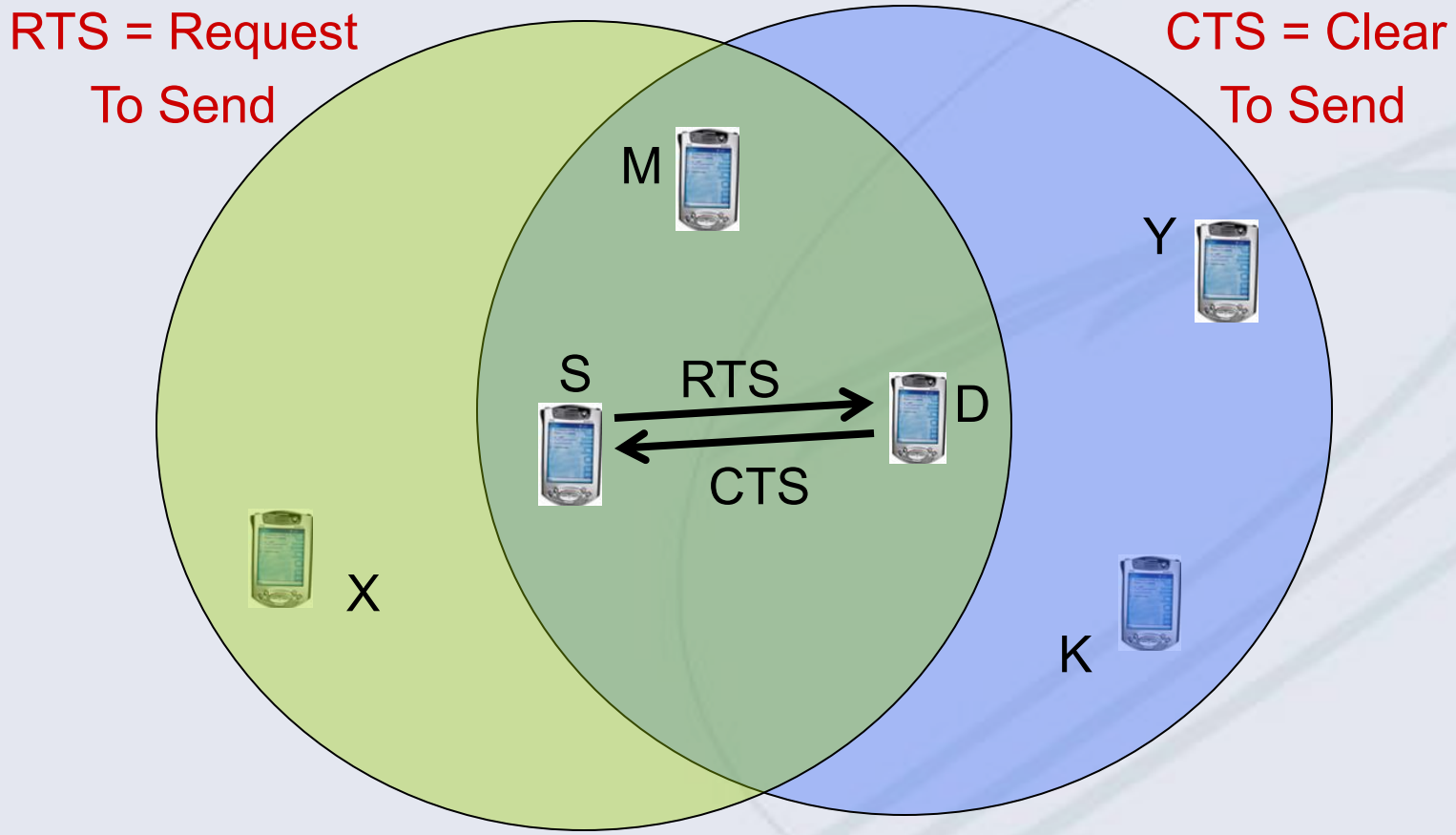
- Transmit and pray
  - Plenty of collisions --> poor throughput at high load
- Listen before you talk
  - Carrier sense multiple access (CSMA)
  - Defer transmission when signal on channel

Wireless Media Disperse Energy

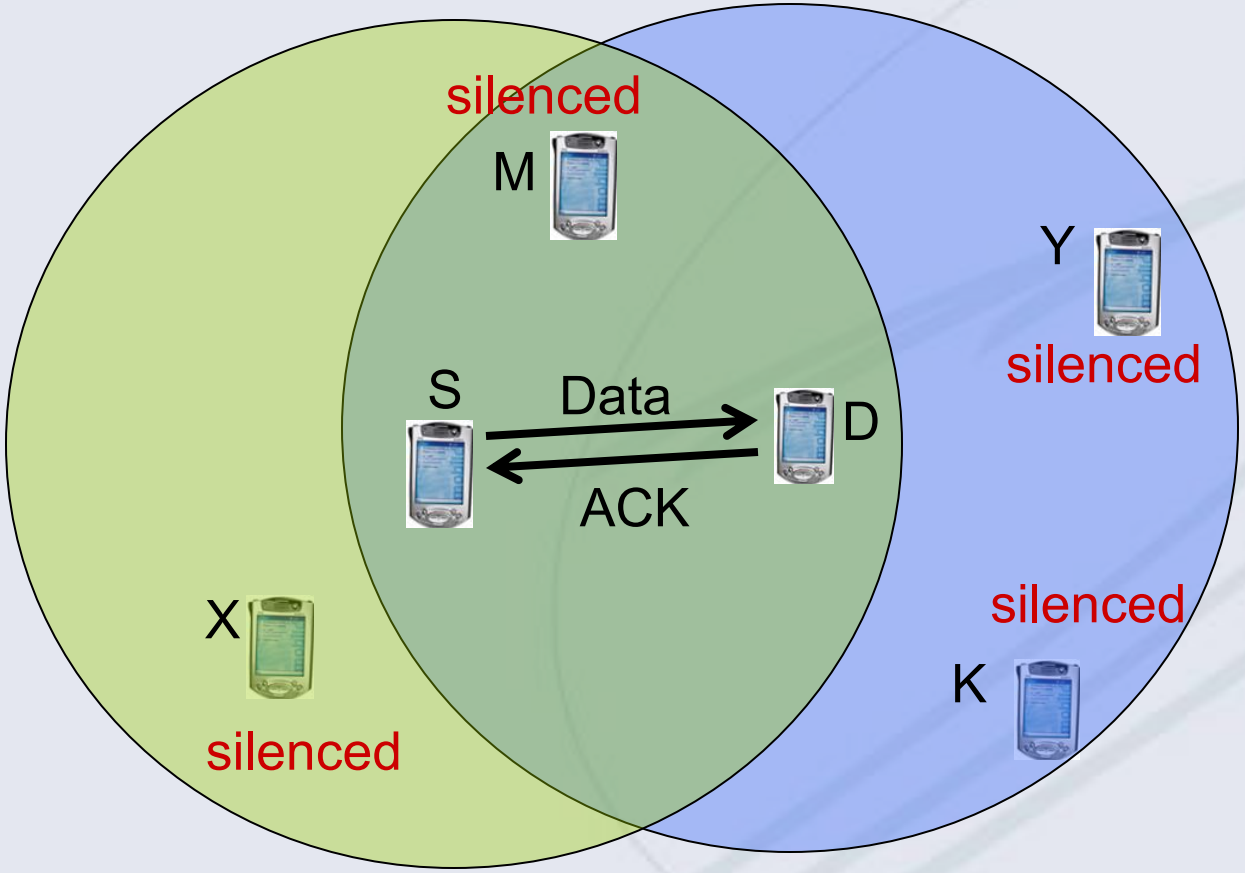
A cannot send and listen in parallel



IEEE 802.11



IEEE 802.11



Wi-Fi generations

Name	IEEE	Certified	Max rate (theory)	Bands
Wi-Fi 4	802.11n	2009	600 Mbit/s	2.4 + 5 GHz
Wi-Fi 5	802.11ac	2013	6.9 Gbit/s	5 GHz
Wi-Fi 6	802.11ax	2019	9.6 Gbit/s	2.4 + 5 GHz
Wi-Fi 6E	802.11ax	2021	9.6 Gbit/s	+ 6 GHz
Wi-Fi 7	802.11be	2024	~23 Gbit/s	2.4 + 5 + 6 GHz
Wi-Fi 8	802.11bn	~2028	focus: reliability	1 - 7.25 GHz

- Wi-Fi 7 is now mainstream: over half of new access points sold in Q2 2026
- Real devices reach a few Gbit/s - the table shows theoretical peaks

Wi-Fi for IoT: what changed recently



Target Wake Time

Wi-Fi 6 sensors agree a schedule and sleep for hours



Multi-Link (Wi-Fi 7)

several bands at once: steadier links for industry



Wi-Fi sensing (802.11bf)

2025 standard: Wi-Fi itself detects presence and motion



6 GHz in Israel

lower 6 GHz band (5925-6425 MHz) open for Wi-Fi 6E / 7

Wi-Fi is the #1 IoT connection type: 32% of all IoT devices (2025).

Wi-Fi HaLow (802.11ah): Wi-Fi for long range

~1 km	sub-GHz	8,191	16 km
typical range	passes walls	devices per access point	record video call (2024)

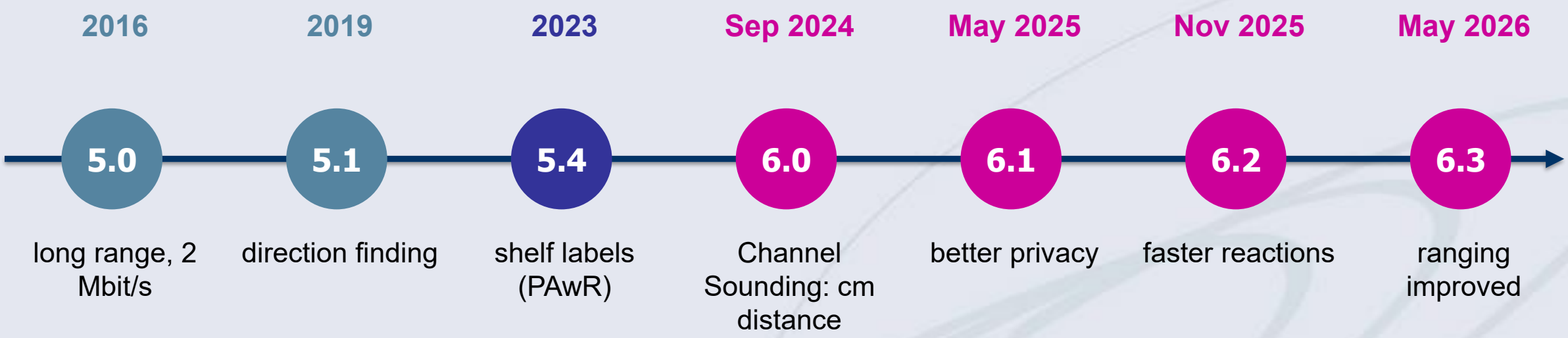
- Normal Wi-Fi security and IP - but below 1 GHz and with low power
- Commercial deployments started in 2025; modules in mass production in 2026
- Sits between Wi-Fi and LoRaWAN: more data than LoRa, more range than Wi-Fi
- Frequencies differ by country - check local rules before buying

What happens when a device joins Wi-Fi

1	**Scan**	find the access point
2	**Associate**	join it
3	**WPA2 / WPA3**	agree on encryption keys
4	**DHCP**	get an IP address
5	**DNS**	find the server
6	**TCP + TLS**	secure connection
7	**MQTT**	start sending data

- Seven steps before the first reading leaves the device
- Each step can fail - and each failure looks different in Wireshark
- A sleeping device repeats much of this when it wakes: Wi-Fi is power-hungry

Bluetooth keeps evolving



5.4 B

Bluetooth devices shipped in 2025

8 B+

per year expected by 2030

Auracast

public broadcast audio is spreading

Bluetooth LE: advertise, connect, read

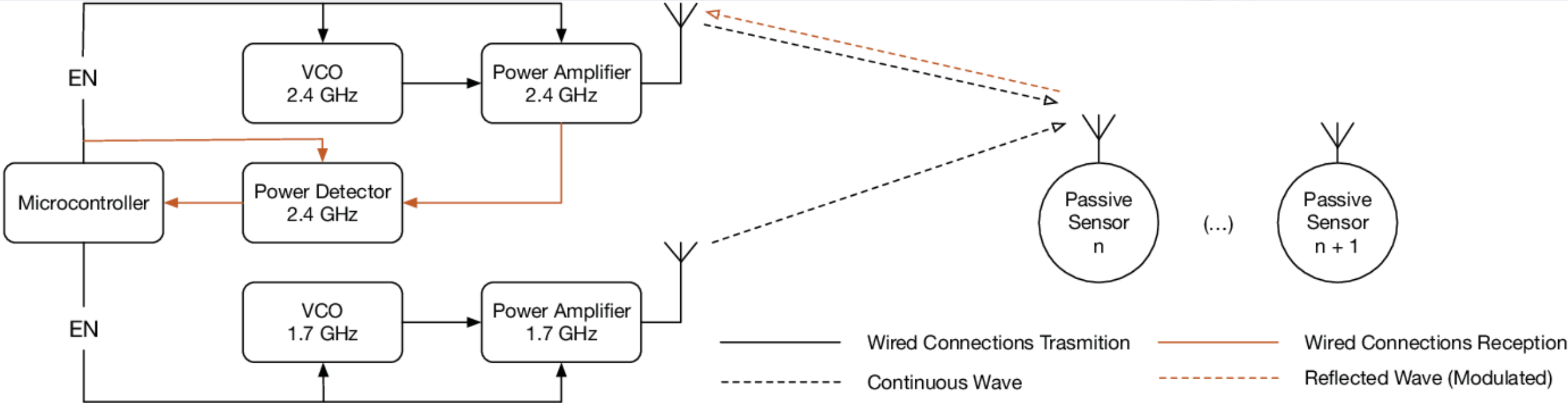


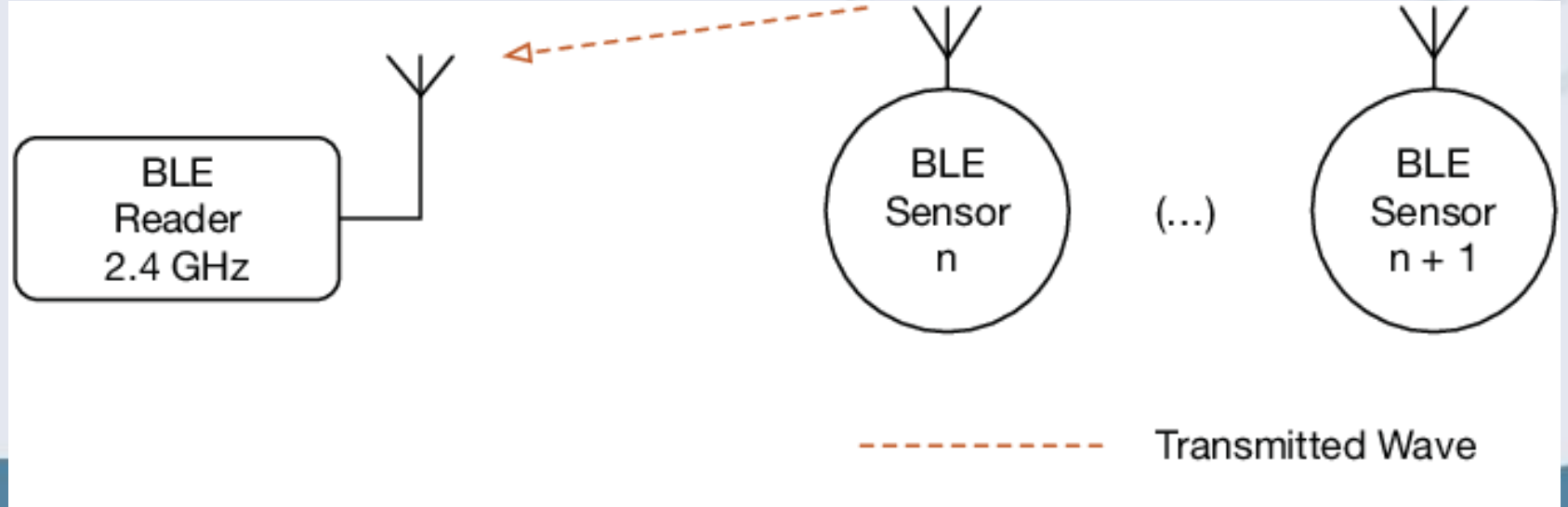
Data on the device

- Temperature service
  - temperature = 23
- Accelerometer service
  - x, y, z
- UART service
  - text in / out

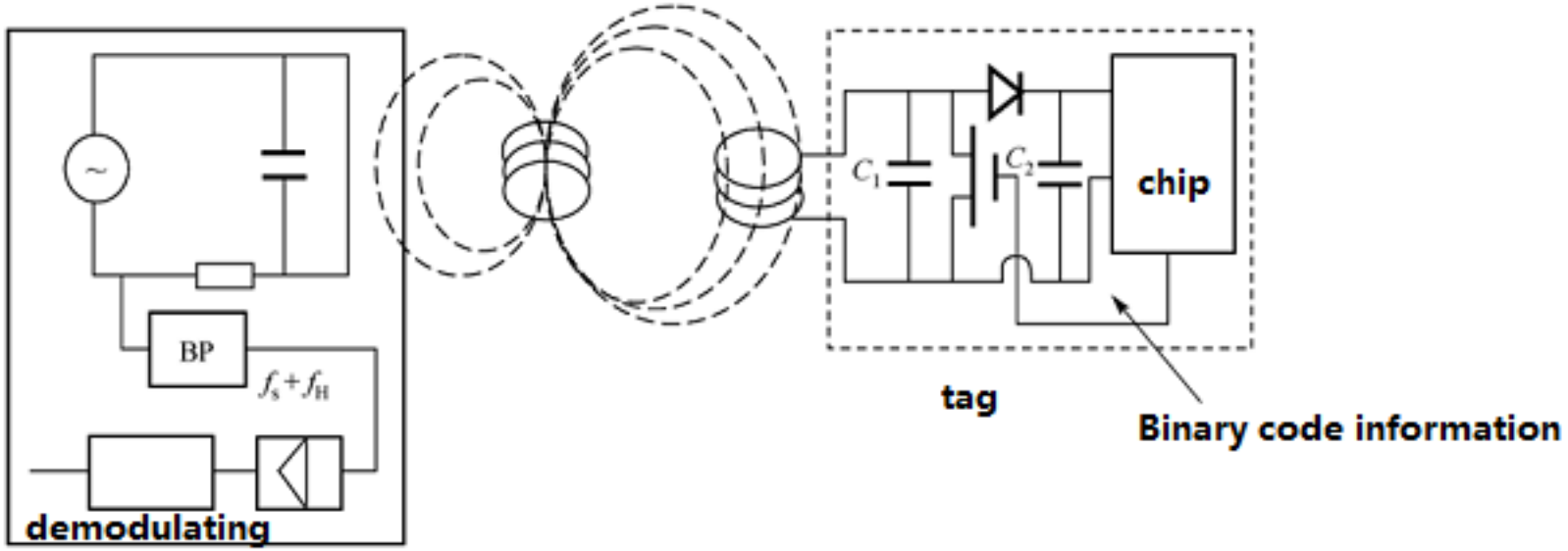
- **Advertising** = shouting your name; **connection** = a private, hopping conversation
- Very low power: the radio sleeps most of the time; range about 10-100 m

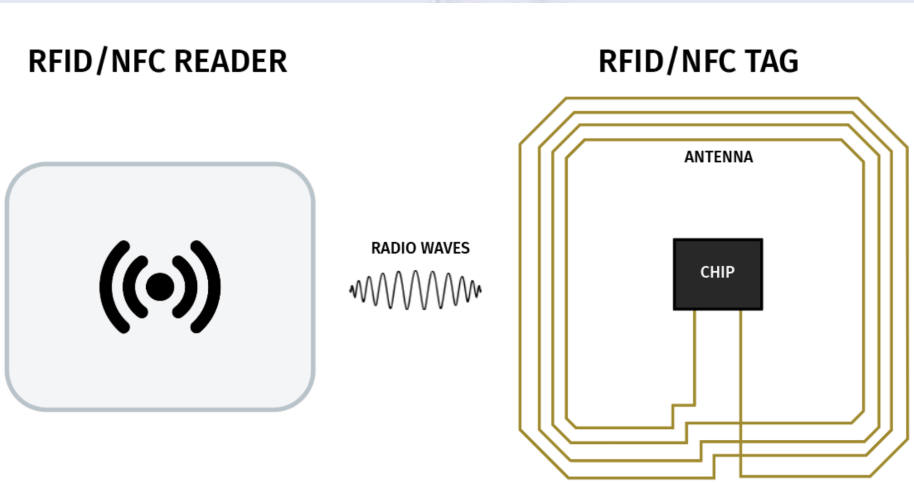
Passive Vs. Active





RFID and NFC: powered by the reader







NFC, RFID and battery-free 'ambient' IoT

0.5 -> 2 cm

NFC Release 15 (June 2025)

52.8 B

RAIN RFID chips shipped in 2024

< 10 cents

battery-free Bluetooth tag

- **Ambient IoT**: tags with no battery, powered by surrounding radio waves
- Walmart (Oct 2025): millions of battery-free sensors in its supply chain
- Being standardized now: 3GPP Releases 19-20 (cellular) and IEEE 802.11bp (Wi-Fi, ~2028)
- NFC roadmap (Feb 2026): up to 8x faster data and post-quantum security

UWB: centimetre-accurate location



- Very short pulses (6-8 GHz); the receiver measures their flight time
- Distance and direction to about 10 cm - far better than signal strength
- Car keys and door locks open only when the phone is really next to them

Aliro 1.0 - Feb 2026	FiRa Core 4.0 - Dec 2025	AirTag 2 - Jan 2026
one phone credential for door locks: NFC, BLE and UWB	the current UWB spec; adds asset tracking	second-generation UWB chip, 1.5x finding range

Competition: Bluetooth Channel Sounding (improved again in 6.3, May 2026) also measures distance at cm level

Wi-Fi and BLE in Wireshark: same idea

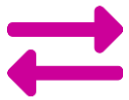
	Wi-Fi capture	Bluetooth LE capture
1	radio info (channel, signal)	radio info (channel, RSSI)
2	802.11 frame (MAC)	Link Layer
3	IP	-
4	TCP / UDP	L2CAP, ATT
5-7	TLS, MQTT, DNS ...	GATT values

BLE carries no IP - a gateway is needed to reach the Internet.

Day 1 in six sentences



Layers let us swap radios without rewriting the application.



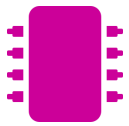
Devices dial out and keep a session alive.



TCP buys reliability; UDP buys speed and battery.



TLS gives privacy, integrity and identity.



Sensors reach the controller via GPIO, UART, I2C, SPI.



Every radio trades range, data rate and power.

Quick quiz

1. The device reaches the cloud, but the cloud cannot reach the device. Why?
2. Five sensors, two wires, each with an address. Which interface?
3. Which Wireshark filter shows only name look-ups?
4. Why does 868 MHz usually reach farther than 2.4 GHz?
5. What three things does TLS give you?
6. Which layer chooses between TCP and UDP?

Thank You!!!